

16th Brazilian-German Frontiers of Science and Technology Symposium

AI for Good: Global Responsible Innovation and Business Transformation

24 – 27 September 2026 | Goiânia, Brazil



Alexander von
HUMBOLDT
STIFTUNG



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WELCOME

01

**Foreword by
conference co-chairs**

We are delighted to welcome you to the 16th Brazilian-German Frontiers of Science and Technology Symposium - BRAGFOST 2026 - hosted this year in the vibrant and welcoming city of Goiânia, Brazil. Known for its lush green parks, rich cultural heritage, and warm hospitality, Goiânia is also rapidly emerging as a dynamic hub for artificial intelligence and deep tech innovation in the heart of the country, making it the perfect setting for our discussions.

AI for Good: Global Responsible Innovation and Business Transformation

Theme of the 16th Brazilian-German Frontiers of Science and Technology Symposium

This year, we come together under the theme **“AI for Good: Global Responsible Innovation and Business Transformation”** to explore how artificial intelligence can contribute to socially beneficial, sustainable, and inclusive futures.

BRAGFOST 2026 brings together outstanding early-career researchers from Brazil and Germany. The symposium offers a space for interdisciplinary exchange across national and disciplinary boundaries, enabling participants to share perspectives, develop networks, and identify new opportunities for international collaboration.

Artificial intelligence is rapidly changing how we work, learn, conduct research, communicate, and make decisions. Its growing presence creates significant opportunities for innovation and transformation, but it also raises fundamental questions about the kind of technological futures we wish to create. How can AI strengthen human capabilities rather than replace human judgement? How can its benefits be made accessible across different sectors, regions, and communities? And how can innovation be aligned with social responsibility, environmental sustainability, and democratic values?

This year's sessions approach these questions from complementary perspectives. They examine the conditions under which AI can be developed and applied in ways that are reliable, transparent, fair, and responsive to diverse societal needs. Across the programme, participants will consider the relationship between technological progress and trust: trust in data, in models, in institutions, and in the ways AI-mediated decisions affect individuals and societies.

A central focus is the challenge of translating AI innovation into meaningful societal and economic value. This includes reflecting on how businesses and organizations can engage with AI in ways that are accessible, human-centered, and sustainable, and how research, industry, and policy can work together to ensure that technological advances do not deepen existing inequalities. In this context, AI for good means not only improving efficiency or competitiveness, but also widening participation and supporting more resilient and inclusive forms of development, and ensuring these advancements transition smoothly from the laboratory into real-world applications.

The programme also addresses the foundations of trustworthy AI. As data increasingly shape the performance and social impact of AI systems, questions of quality, representativeness, privacy, security, and accountability become ever more important. The symposium will explore how AI

systems can be designed and assessed in ways that recognise uncertainty, reduce harm, and foster public confidence - particularly in an environment where synthetic content and powerful language models are becoming part of everyday life.

At the same time, AI is transforming the production, communication, and validation of knowledge. In education, research, and public discourse, generative AI challenges established assumptions about creativity, authorship, expertise, and critical judgement. These developments invite us to reconsider how learning and knowledge transfer can remain participatory, diverse, and meaningful, and how educational and research institutions can prepare people not simply to use AI, but to understand, question, and shape it responsibly.

The presentations and discussions are organized around the following topics:

1	Empowering SMEs with AI: Bridging the Innovation Gap
2	Synthetic Data for Trustworthy AI: Foundations, Modalities, and Responsible Development
3	Trustworthy AI in the Age of LLMs
4	Generative AI for Next-Generation Education, Research, and Knowledge Transfer

We warmly thank all participants for their valuable contributions and their willingness to engage in open, critical, and constructive dialogue. We hope that BRAGFOST 2026 will inspire new perspectives, lasting partnerships, and collaborative approaches to advancing AI innovation that is not only technologically capable, but also responsible, equitable, and beneficial to society.

Beyond the technical sessions, we invite you to experience the unique rhythm of Goiânia. Whether it is continuing our debates over a traditional local barbecue, sharing a great specialty coffee, or simply enjoying the city's beautiful parks, we hope the local culture fuels both your creativity and our collaborative spirit.

We wish you an inspiring symposium, fruitful discussions, and an enjoyable time networking in Goiânia.

 **Arlindo Galvão and Evelyn Reinmuth**
Conference Co-Chairs



PRACTICAL INFORMATION

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Conference information

About Brazilian-German Frontiers of Science and Technology Symposium

The Brazilian-German Frontiers of Science and Technology (BRAGFOST) symposia are a series of interdisciplinary conferences co-organized by the Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES) and the Alexander von Humboldt Foundation. Funding on the German side is provided by the Federal Ministry of Research, Technology and Space.

The symposium brings together approx. 60 outstanding young researchers from Brazil and Germany coming from various universities, research institutions, and industry, to introduce their areas of research and technical work, thus enabling an interdisciplinary transfer of knowledge and methods. Researchers working in different fields of sciences and engineering build cross-disciplinary networks and take the opportunity to work on future international collaborations.

The Alexander von Humboldt Foundation supports collaboration between the BRAGFOST participants with its dedicated follow-up program CONNECT which grants residence allowances for working visits in the partner country for up to 30 days.

This year's symposium is dedicated to the theme How to improve model robustness.

The organizing committee consists of two conference co-chairs and six session chairs from both countries. They jointly developed the program for this conference.

Conference venue

The Brazilian-German Frontiers of Science and Technology Symposium 2026 will be held at:

K Hotel, Av. Dep. Jamel Cecílio, 2550 - Jardim Goiás, Goiânia – GO

Throughout the conference period, a block of single hotel rooms has been reserved for all participants at the hotel. To avoid double reservations, please do not contact the hotel directly.

Room charges for all invited participants will be billed directly to the Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES) and the Center of Excellence in Artificial Intelligence (CEIA) for the following dates:

- Arrival 24 September 2026 (or 23 September 2026 for participants from Germany)
- Departure 27 September 2026 (or 28 September 2026 for participants from Germany).

Any additional nights apart from the dates listed will have to be covered by the individual participant.

You will also be responsible for additional charges or any incidental charges upon check-out.

Travel directions

The 16th Brazil-Germany Symposium on Frontiers of Science and Technology (BRAGFOST 2026) will take place at K Hotel Goiânia, located in Goiânia, Goiás, Brazil.

The hotel is situated in the Jardim Goiás neighborhood, one of the city's main commercial and business areas, with convenient access to restaurants, shopping centers, and other amenities.

Getting to the Hotel

K Hotel is located in Jardim Goiás, one of Goiânia's main business and leisure districts, at Avenida Deputado Jamel Cecílio, 2550. The hotel is close to major commercial and cultural attractions, including Flamboyant Shopping Mall, Flamboyant Park and the Oscar Niemeyer Cultural Center.

For participants arriving in Goiânia by air, the closest airport is Santa Genoveva, Goiânia International Airport (GYN), located approximately 15 km from the hotel. The journey from the airport to the K Hotel takes approximately 20 to 30 minutes by car, depending on traffic conditions.

Participants arriving at GYN can reach the hotel by taxi or app-based transportation services. Taxis are available at the airport, and ride-hailing services such as Uber and 99 are widely available in Goiânia. Airport transfers for the participants will be arranged by the event organizers. Further information will be provided to participants in advance.

Hotel info: <https://khotel.com.br/en/>

Maps: <https://maps.app.goo.gl/6NxTFL1STTQ9bCZG7>

Conference language

The official conference language is English. It will be used for all printed materials, presentations, and discussions. Please note that the BRAGFOST partner organizations are not responsible for the content of documents and conference materials submitted by the participants.

Photos

Photos taken during the conference may be published as part of the information and public relations work of the Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES) and the Alexander von Humboldt Foundation. They will not be used for commercial purposes. If you do not agree with the publication of the photos in which you are depicted, you can object by email to comunicacao@ceia.ufg.br at any time.

Exclusion of liability

Please understand that both partner organizations are neither liable for any physical injury or damage to property that may occur during your travels to the meeting site and back home nor during your stay at the meeting site itself.

Code of conduct

The Alexander von Humboldt Foundation requires all attendees, speakers, exhibitors, guests and staff at our conference to agree with the following code of conduct. The Alexander von Humboldt Foundation will enforce this code throughout the entire event.

We expect cooperation from all participants to help ensure a safe environment for everybody.

Our event is designed to provide a respectful environment and harassment-free conference experience for everyone, regardless of gender identity and expression, age, sexual orientation, disability, physical appearance, body size, ethnicity, nationality, or religion (or lack thereof). We do not tolerate harassment of participants in any form.

Sexual language and imagery are not appropriate for any conference venue, including talks, workshops, receptions, or via any online (social) media. Participants violating these rules may be sanctioned or expelled from the conference at the discretion of the Alexander von Humboldt Foundation. Participants experiencing harassment are encouraged to contact a staff member. Staff can be identified by badges or contacted via email (stephanie.dill@avh.de). Conference staff will be happy to help or assist those experiencing harassment to feel safe for the duration of the conference. We value your attendance.



PROGRAMME

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Conference schedule

Thursday, 24 September 2026

from 14:00	Check in (K Hotel Goiânia)
18:00 – 19:00	Session coordination meeting (<i>session chairs & speakers only</i>)
19:00 – 20:00	Speed meeting
20:00	Welcome remarks & Dinner

Friday, 25 September 2026

09:00 – 09:15	Opening remarks
09:15 – 10:30	Session I - <i>Synthetic Data for Trustworthy AI: Foundations, Modalities, and Responsible Development</i>
10:30 – 11:00	Group photo + break
11:00 – 12:15	Session I (cont.)
12:15 – 13:15	Lunch
13:15 – 13:45	Presentation of CAPES + AvH funding opportunities
13:45 – 14:45	Poster flash talk presentations
14:45 – 15:45	Poster session I (odd numbers)
15:45 – 16:00	Break
16:00 – 17:00	Poster session II (even numbers)
17:00 – 18:00	Workshop I: Identification of potential collaboration partners and projects
19:00	Dinner

Saturday, 26 September 2026

09:00 – 10:15	Session II - <i>Trustworthy AI in the Era of Large Language Models (LLM)</i>
10:15 – 10:30	Break
10:30 – 11:45	Session II (cont.)
11:45 – 12:45	Lunch
12:45 – 14:00	Session III - <i>Empowering SMEs with AI: Bridging the Innovation Gap</i>

14:00 – 14:15	Break
14:15 – 15:30	Session III (cont.)
16:00	Beginning of cultural tour
19:00	Dinner at a local restaurant
21:00	Return to the hotel

Sunday, 27 September 2026

by 12:00	Check out
09:00 – 10:15	Session IV: <i>Generative AI (Gen AI) for Next Generation Education and Research</i>
10:15 – 10:30	Break
10:30 – 11:45	Session IV (cont.)
11:45 – 12:45	Workshop II: Position paper on <i>Responsible AI Collaboration for SMEs and Society - 2026</i>
12:45 – 13:00	Wrap-up & Closing remarks
13:00	Farewell lunch
thereafter	Departure



WHO IS BEHIND IT

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Organizing committee

The programme of BRAGFOST 2026 was jointly developed by two conference co-chairs and eight session chairs from Brazil and Germany.

CONFERENCE CO-CHAIRS

Arlindo Galvão	Federal University of Goiás
Evelyn Reinmuth	University of Hohenheim

SESSION CHAIRS

Synthetic Data for Trustworthy AI: Foundations, Modalities, and Responsible Development

Thiago Naves	Federal University of Technology Parana
Denilton Darold	Fraunhofer ISI

Trustworthy AI in the age of LLMs

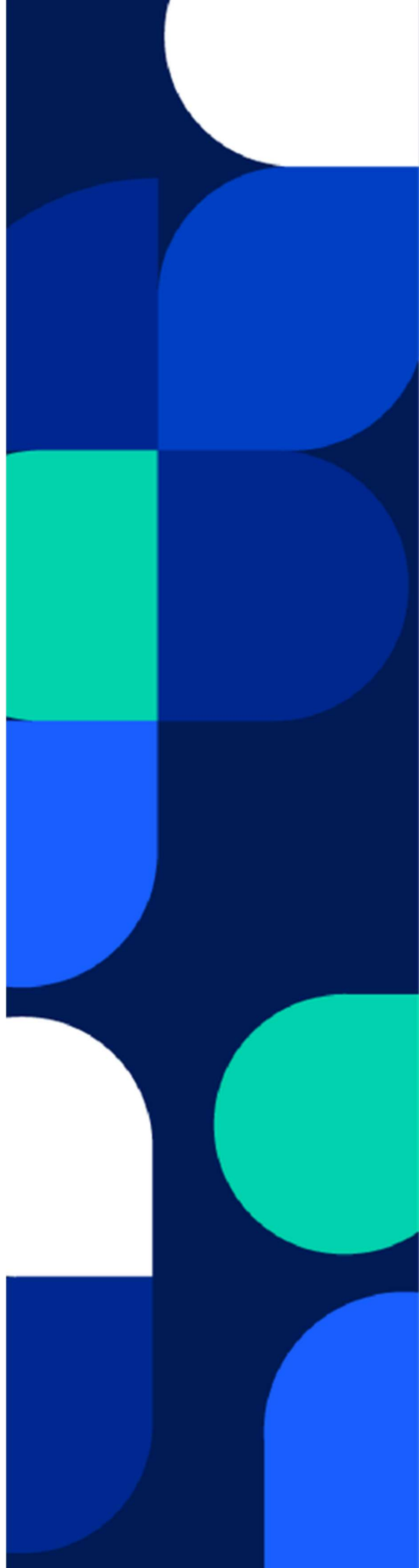
Aline Paes	Federal Fluminense University
Sabine Weber	University of Bamberg

Empowering SMEs with AI: Bridging the Innovation Gap

Rodrigo Zempulski Fanucchi	Federal University of Goiás
Michael Dunst	Esslingen University of Applied Sciences

Generative AI for Next-Generation Education, Research, and Knowledge Transfer

Carlos Vinicius Gomes Melo	University of Sao Paulo
Oliver Wiczorek	University of Kassel



SESSIONS AT A GLANCE

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List of sessions

Synthetic Data for Trustworthy AI: Foundations, Modalities, and Responsible Development

Session chairs Thiago Naves, Federal University of Technology Parana & Denilton Darold, Fraunhofer ISI

Beyond Augmentation: Using Synthetic Data Responsibly	Isabel Huebener, Deggendorf Institute of Technology
Can Synthetic Data Make AI Fairer?	Mariana Recamonde Mendoza, Federal University of Rio Grande do Sul
Toward Data-Centric and Trustworthy AI for Software Engineering	Binger Chen, HU Berlin
Synthetic Realities and the Challenge of Digital Trust	Leopoldo Andre Dutra Lusquino Filho, Sao Paulo State University / University of Campinas

Trustworthy AI in the age of LLMs

Session chairs Aline Paes, Federal Fluminense University & Sabine Weber, University of Bamberg

Conceptual Clarity for Trustworthy AI	Vagrant Gautam, Heidelberg Institute for Theoretical Studies
Operationalizing Explainability: What Legal and Technical Requirements Reveal about the Limits of LLM Interpretability	Aline Macohin, Pontifical Catholic University of Parana
Bridging Interpretability and Alignment for Trustworthy Large Language Models	Francielle Vargas, University of Chile
Beyond the Global North: Mapping Geographic and Geocultural Biases in Trustworthy AI	Anna Kruspe, Munich University of Applied Sciences

Empowering SMEs with AI: Bridging the Innovation Gap

Session chairs Rodrigo Zempulski Fanucchi, Federal University of Goias & Michael Dunst, Esslingen University of Applied Sciences

Understand to adopt: the importance of making AI eXplainable	Kevin Bauer, Goethe University Frankfurt
Human and Data-Driven Intelligence in Technology Roadmapping	Maicon Oliveira, University of Sao Paulo
Architecting Human-AI Symbiosis: Designing Virtual Worlds for Collaborative Performance	Alyson Matheus de Carvalho Souza, Federal University of Rio Grande do Norte
AI-Enabled Precision Livestock Farming in Integrated Crop-Livestock Systems: Current Applications and Future Challenges at the Gladbacherhof Research	Deise Knob, University of Giessen

Generative AI for Next-Generation Education, Research, and Knowledge Transfer

Session chairs Carlos Vinicius Gomes Melo, University of Sao Paulo & Oliver Wieczorek, University of Kassel

From Output to Agency: What Creativity Can Teach Us About Education in the Age of Generative AI	Caroline Menezes, BSP Business & Law School Berlin
Who Shapes Knowledge in the Age of Generative AI? Rethinking Peer Review, Publishing, and Global Equity in Next-Generation Research	Felix Holl, Neu-Ulm University of Applied Sciences
Developing Critical AI Literacies (CAIL) for Sociodigital Futures in Education: A Multidimensional Model	Priscila Gonsales, University of Campinas
Child Rights by Design in AI applied to Education	Rodrigo Nejm, Alana Institute, Sao Paulo



DEEP DIVE

06

Session and presentation abstracts

Synthetic Data for Trustworthy AI: Foundations, Modalities, and Responsible Development

Synthetic data is becoming a key enabler of trustworthy artificial intelligence. Beyond addressing data scarcity and privacy constraints, it supports more robust, fair, and reproducible AI systems while facilitating regulatory compliance. At the same time, its growing adoption raises challenges related to data fidelity, bias propagation, evaluation, digital authenticity, and responsible use.

This session explores the role of synthetic data in trustworthy AI, bringing together researchers from Brazil and Germany to discuss responsible data generation, data-centric AI, fairness-aware machine learning, digital forensics, and trustworthy software engineering. Together, these perspectives show how synthetic data can accelerate AI innovation while addressing challenges of transparency, security, fairness, and governance.

Through expert presentations and discussion, participants will examine the technical, ethical, and societal implications of synthetic data, fostering dialogue on its responsible integration into AI development to ensure reliability, accountability, and public trust.

Key themes:

- **Responsible Synthetic Data Generation:** Opportunities and limitations of synthetic data, including data quality, fidelity, transparency, human oversight, and evaluation strategies for responsible AI.
- **Data-Centric and Trustworthy AI:** Integrating synthetic and real-world data, retrieval-based methods, and structured knowledge to improve reliability, robustness, and reproducibility in AI systems.
- **Fairness and Robustness through Synthetic Data:** Using synthetic data to mitigate representation bias, improve data diversity, enable stress testing, and support more equitable machine learning models.
- **Security, Digital Trust, and Governance:** Addressing synthetic media detection, digital forensics, misinformation, privacy preservation, and regulatory challenges, including GDPR and LGPD.

Beyond Augmentation: Using Synthetic Data Responsibly

Isabel Huebener, Deggendorf Institute of Technology

Artificial intelligence systems increasingly depend on large amounts of data, yet high-quality and representative datasets remain scarce due to privacy constraints, cost, and limited availability. Synthetic data—artificially generated information designed to mimic real-world statistics—has emerged as a promising solution. It enables data augmentation, supports the simulation of rare scenarios, and opens new possibilities for experimentation across domains such as healthcare, agriculture, and industrial systems.

However, synthetic data is not simply a drop-in replacement for real data. This talk focuses on a central question: when does synthetic data improve AI systems—and when does it quietly degrade them?

I will present a concise, application-driven perspective on synthetic data pipelines and highlight two critical challenges. First, bias amplification and loss of fidelity: synthetic data often inherits and reinforces patterns from the original data, while failing to preserve complex relationships in the real world. Second, the evaluation paradox: models trained on synthetic data are increasingly evaluated on synthetic benchmarks, creating a circular validation loop that can lead to overestimated performance and reduced robustness.

Based on these observations, the talk introduces a simple practical framework for responsible use of synthetic data in applied settings:

1. Use synthetic data selectively, targeting specific weaknesses (e.g., rare or under-represented cases) rather than replacing entire datasets.
2. Maintain human oversight, integrating domain expertise to validate generated data and detect artifacts or unrealistic patterns.
3. Ensure transparency, documenting how synthetic data are generated and used within the pipeline.

Rather than providing a purely technical overview, this talk aims to bridge perspectives between research and practice. It invites discussion on a key challenge for the next generation of AI systems: how to balance scalability and data availability with reliability and trust.

Can Synthetic Data Make AI Fairer?

Mariana Recamonde Mendoza, Federal University of Rio Grande do Sul

Machine learning (ML) models increasingly support critical decisions across domains such as finance, criminal justice, education, and healthcare. In these high-stakes settings, predictive systems often influence outcomes such as credit approval, risk assessment, admission decisions, and access to specialized services, directly shaping the allocation of resources, opportunities, and benefits. However, real-world datasets frequently exhibit substantial class imbalance and reflect historical and structural inequalities. As a result, trained models may systematically underperform for underrepresented or historically marginalized populations, potentially perpetuating or amplifying existing disparities.

Performance differences across demographic groups are often associated with algorithmic bias, although not all observed disparities necessarily constitute unfairness in a normative sense. In ML, bias can arise from multiple sources, including imbalanced data, measurement errors, and modeling decisions, becoming a fairness concern when it leads to unjustified differences in outcomes. Such issues may emerge at any stage of the AI development lifecycle and disproportionately affect certain populations in high-impact decision-making contexts.

Recent research has increasingly examined the relationship between fairness and synthetic data, highlighting both challenges and opportunities. On one hand, biases encoded in generative models may reproduce or even intensify existing inequalities. On the other, synthetic data offers a promising approach for mitigating representation-related biases and improving data diversity. It also enables systematic stress testing of predictive models under underrepresented conditions, facilitating the simulation of edge cases and vulnerabilities that might otherwise remain undetected during model development and evaluation.

This talk explores the dual role of synthetic data in fairness-aware AI, examining both its potential to support bias mitigation and algorithmic equity and the technical and ethical challenges involved in realizing this potential. Through this perspective, we investigate whether and how synthetic data can contribute to more trustworthy and equitable AI systems by addressing systemic sources of bias, while highlighting key open questions for responsible AI development, particularly regarding fairness and the prevention of discriminatory outcomes.

Toward Data-Centric and Trustworthy AI for Software Engineering

Binger Chen, HU Berlin

The rapid development of generative AI and large language models is changing software engineering, advancing automation in tasks such as code comprehension, generation, and translation. Meanwhile, their use in practice raises reliability concerns. Generated code can contain subtle errors or inconsistencies, which can easily lead to incorrect behavior and functionality in practice. This becomes particularly challenging when models rely heavily on synthetic data that does not always reflect real-world development scenarios.

A line of work I have been involved in focuses on leveraging large-scale code repositories as an alternative source of knowledge for these tasks. Earlier work explored cross-language code retrieval and data augmentation methods that reuse existing code instead of generating it from scratch. The underlying idea is that human-written code already captures many constraints of programming languages, and reusing such artifacts can help maintain syntactic correctness and more realistic behavior. In this setting, retrieval is not only used as a lookup mechanism but combined with learned representations and user feedback to iteratively improve results.

More recently, this perspective has been extended to combining large language models with structured information and interactive workflows. One example from our research is the use of curated databases of foundation models together with LLM-based agents that assist users in selecting suitable models for a given task. These systems rely on real-world metadata and allow users to refine requirements through interaction, making the decision process more reproducible and trustworthy than relying solely on automated AI systems.

The talk connects these lines of work to discuss trustworthy AI in software engineering. In particular, it considers how real-world data and structured knowledge can complement generative AI, how interaction can help mitigate errors, and where current approaches still face limitations. This also raises questions about how to combine retrieval and generation more effectively, how to evaluate correctness beyond standard benchmarks, and how to design systems that remain reliable when deployed in practice.

Synthetic Realities and the Challenge of Digital Trust

Leopoldo Andre Dutra Lusquino Filho, Sao Paulo State University / University of Campinas

Recent advances in generative artificial intelligence have made synthetic images, voices, videos, and texts increasingly realistic and accessible. While these technologies enable important innovations in areas such as education, healthcare, accessibility, and software development, they also raise new challenges for digital trust. As synthetic content becomes harder to distinguish from authentic media, questions emerge about misinformation, fraud, scientific integrity, and the reliability of digital evidence itself.

This talk discusses synthetic realities from the perspective of digital forensics and media detection. Drawing on ongoing work within the Horus Project at the University of Campinas, the presentation examines how contemporary forensic approaches attempt to identify residual signals associated with AI-generated content, including inconsistencies related to acquisition pipelines, statistical regularities, multimodal coherence, and contextual provenance. The talk argues that recent advances in generative models challenge some of the traditional assumptions of multimedia forensics, particularly the idea that manipulations necessarily leave stable and detectable artifacts. In increasingly adversarial and open-set scenarios, the problem shifts from merely identifying isolated artifacts toward reasoning about context, uncertainty, and cross-modal consistency.

The presentation introduces some of the current challenges faced by forensic AI systems, including rapidly evolving generative models, limited access to reliable training data, and the difficulty of generalizing detection methods to previously unseen synthetic media. Examples involving misinformation, manipulated scientific images, phishing campaigns, and AI-generated multimodal content will be used to illustrate these challenges in accessible terms.

By connecting technical limitations with broader societal implications, the talk aims to discuss synthetic realities not only as a detection problem, but also as an emerging epistemological challenge for computational systems that mediate knowledge, communication, and evidence. In this setting, trustworthy AI depends not solely on increasingly accurate classifiers, but also on interpretability, contextual reasoning, provenance analysis, and interdisciplinary approaches capable of addressing the evolving relationship between synthetic generation and digital authenticity.

Trustworthy AI in the age of LLMs

The session will examine a set of interrelated technical and societal questions aimed at deepening understanding of LLM behavior and its broader implications. In doing so, we will trace the full life cycle of the model, from data collection and annotation through model design and training to evaluation and deployment.

We will start out by defining key terms of the discussion exploring how abstract concepts—such as fairness, safety and responsibility—are operationalized and what “trustworthy” and “AI” can mean in this context. Next, we discuss how toxic or harmful behaviors relate to model internals, exploring the extent to which biases and undesirable outputs can be linked to representational and training dynamics, interrogating the boundaries between explainability methods and legal requirements.

A central premise is that understanding why models behave the way they do is a necessary first step toward effective intervention, governance and long-term sustainability. How can evaluation protocols either help to discover bias or further enforce it? How can calibrated confidence help users to make informed decisions about model output? And what are the cultural and linguistic nuances in observed harmful model behavior?

In addition to these methodological points, the discussion will further consider the impact of interpretability research on natural language processing as a whole, highlighting how advances in this area can inform evaluation, debugging, regulatory compliance, and accountability practices.

The specific topics to be discussed during the session include:

- Dataset and annotation artifacts as sources of harm
- Explaining and tracing hallucinations
- Relating toxic behavior to model internals
- Evaluation protocols for interpretability and safety
- Uncertainty estimation and calibrated confidence in LLM outputs
- Cultural and linguistic variation in harmful behavior
- Understanding the role of smaller and more sustainable language models in harm mitigation and alignment
- Examining how abstract concepts are operationalized in the models
- Impact of interpretability research on NLP

Conceptual Clarity for Trustworthy AI

Vagrant Gautam, Heidelberg Institute for Theoretical Studies

As participants in a session about trustworthy AI in the age of LLMs, can we even agree on what “trustworthy,” “AI,” and the “age of LLMs” actually mean? I argue that conceptual clarity, or deeply and critically interrogating the meanings of such concepts is critical to understanding and achieving the goal of trustworthy AI in the age of LLMs. Behind the veneer of advertising hype from big tech companies, the age of LLMs is characterized by extraction, theft, pollution of natural and information ecosystems, devaluation of skills, expertise, and labour, and the concentration of power and capital in the hands of a few. In such an age, what can “trustworthy AI” possibly look like? “AI” is also a term that is greatly overloaded and can benefit from more specificity - sometimes it is large language models generating text, sometimes it is image generation tools, and sometimes it is domain-specific machine learning applications trained to identify abnormalities in medical scans. Disentangling the many faces of “AI” is a key part of the conceptual clarity we need in order to rigorously audit specific systems in specific contexts for actual benefits, to decide when we want them and when we do not. For “trustworthy” machine learning applications, in turn, we can use inspiration from the social sciences, where concepts such as trust and credibility have been studied for decades before LLMs came to the fore. System designers should take inspiration from characteristics of trust such as fairness, benevolence, honesty, and competence. Such abstract characteristics need clear definitions and should be operationalized multidimensionally, highlighting that conceptual clarity is key at all levels to developing trustworthy machine learning applications in this age.

Operationalizing Explainability: What Legal and Technical Requirements Reveal about the Limits of LLM Interpretability

Aline Macohin, Pontifical Catholic University of Parana

Interpretability research has produced an extensive body of methods, yet a fundamental question remains open: under what criteria does an explanation qualify as valid? In the absence of a consensual answer within the literature, regulatory regimes and technical-normative standards have begun to fill this gap, frequently establishing requirements that presuppose levels of explainability, contestability, and auditability not yet fully supported by available methods. In the Brazilian context, this landscape is shaped by Article 20 of the General Data Protection Law (LGPD), Bill 2338/2023 (currently pending before the National Congress, inspired by the EU AI Act), the constitutional duties of reason-giving and publicity in public decision-making, and the IEEE 7001-2021 standard, which is distinctive in operationalizing transparency as a set of graduated, testable levels of compliance for autonomous systems.

This work critically examines the relationship between such requirements and the actual capabilities of interpretability techniques, analyzing the operationalization of concepts such as fairness, accountability, and explainability within model architectures and evaluation protocols. Three systematic mismatches are identified: the low faithfulness of post-hoc explanations; the degradation of uncertainty calibration under distribution shift; and the insufficiency of benchmarks to capture culturally and linguistically situated harms, particularly in Portuguese.

As an exemplary technical pathway for addressing these mismatches, the work discusses prediction with abstention (also referred to as selective prediction): a mechanism in which the model conditions its response on a measurable confidence threshold and, below that threshold, abstains and returns the decision to a human reviewer. Rather than resolving the validity question, abstention substitutes an auditable guarantee of allocation — when the system answers and when it defers — for an unattainable guarantee of valid explanation. We argue that it operationalizes the relevant normative requirements in testable but uneven form. It maps most directly onto the right to review under Article 20 of the LGPD — which no longer expressly requires that the review be performed by a natural person, so that in public-sector decision-making the duty of human review and justification rests primarily on the constitutional duties of reason-giving and due process. Its correspondence to the duty to give reasons is only partial, since above the threshold the system still decides without justifying, and its alignment with IEEE 7001-2021 holds at the level of documenting when the system defers rather than why it decides. This articulation is itself bounded by the second mismatch above: because abstention depends on the model's confidence estimate, miscalibration under distribution shift can suppress it precisely on the cases where human review is most warranted.

Finally, the paper addresses the risks of transparency washing and argues for broadening the empirical and linguistic diversity considered in the development and evaluation of AI systems — particularly for Portuguese — as a condition for their robustness, reliability, and suitability across deployment contexts.

Bridging Interpretability and Alignment for Trustworthy Large Language Models

Francielle Vargas, University of Chile

Large Language Models (LLMs) are increasingly deployed in high-stakes applications, raising concerns about transparency, reliability, and alignment with human values. While interpretability aims to uncover the internal mechanisms behind model decisions, and alignment focuses on shaping model behavior, these directions have largely evolved in isolation. In this talk, I present a unified approach that bridges interpretability and alignment by using human rationales as supervision signals for model internals. I introduce methods that align attention mechanisms with human-annotated rationales, enabling models to become self-explaining while improving behavioral reliability. This approach turns interpretability into an intrinsic mechanism for actively guiding model behavior. I demonstrate the effectiveness of this framework across tasks such as harmful content detection and retrieval-augmented generation (RAG), where aligning model attention with relevant evidence improves both faithfulness and plausibility of outputs. Finally, I discuss limitations, including challenges in evaluating faithfulness and risks of superficial alignment, and outline directions for developing more robust and trustworthy LLMs.

Beyond the Global North: Mapping Geographic and Geocultural Biases in Trustworthy AI

Anna Kruspe, Munich University of Applied Sciences

What is the research about? While the discourse on "Trustworthy AI" often focuses on technical robustness or personal data privacy, my research addresses a more subtle but equally critical dimension of fairness: geographic and geocultural representation.

Generative AI and Large Language Models (LLMs) are not "placeless" systems; they are trained on datasets that heavily over-represent the Global North. My work investigates how these models perceive different regions of the world, how they reproduce regional stereotypes (e.g., within Germany), and how they struggle to capture the nuances of non-Western cultures in both text and multimodal outputs like music.

Why does it matter? If AI is to be "for Good" and foster "Global Responsible Innovation," it must be inclusive. A model that "hallucinates" the geography of the Global South or reduces complex cultural identities to clichés is not trustworthy. For SMEs and educational institutions in countries like Brazil, relying on models with a heavy "West-centric" bias can lead to a loss of cultural sovereignty and the marginalization of local knowledge. In a globalized world, "Trustworthy AI" must mean "Culturally Representative AI."

What are the most exciting results of the past several years? My recent findings show that the "intelligence" of LLMs is highly location-dependent. For instance, models exhibit a significantly higher factual density for Western urban centers compared to rural or Global South regions. Furthermore, we discovered a "linguistic-cultural bottleneck": when prompted in English, generative music models often produce high-quality but homogenized results, whereas prompts in local languages or for specific regional genres often trigger Westernized "clichés" or lower-quality outputs. This confirms that the bias is not just in the language, but in the underlying cultural mapping of the models.

What are the current limitations and major challenges? The biggest challenge lies in the data hegemony of English-language content, which fosters an "algorithmic monoculture." This leads to cultural erasure, where localized knowledge and diverse artistic expressions are flattened into a global, Western-centric average.

Furthermore, the intersectional nature of these biases—where geographic origin overlaps with gender or socio-economic status—is still poorly understood. Technically, LLMs remain "black boxes," making it difficult to pinpoint exactly which training data or model components are responsible for a specific biased output.

What should the future hold? The future of Trustworthy AI must be built on three pillars:

Interpretability & XAI: We must develop methods to identify the specific "neurons" or layers responsible for geoculturally biased behavior, allowing for targeted "surgical" corrections rather than blunt filters.

Sovereign Personalization: While personalization offers a path to cultural relevance, it must be designed carefully to avoid creating "cultural echo chambers." We need systems that empower users to define their own cultural context without sacrificing global connectivity.

Critical AI Literacy: A major part of the solution is visibility. We must foster a global awareness that AI is not a neutral authority. By making these hidden biases visible, we can move from blind trust to a critical, informed engagement with generative systems.

Empowering SMEs with AI: Bridging the Innovation Gap

Small and medium-sized enterprises (SMEs) are central drivers of economic growth and employment in both Brazil and Germany. Despite this importance, many SMEs struggle to adopt artificial intelligence due to limited resources, lack of knowledge about where to use and expertise about how to implement, and uncertainty about ethical, legal, and organizational implications. The objective of this session is to explore how responsible, accessible, and scalable AI solutions can empower SMEs to overcome these barriers and actively participate in global, AI-driven value chains.

The session aims to bridge perspectives from research, industry, and policy, highlighting how AI can be designed and deployed not only for efficiency and competitiveness, but also for inclusiveness, sustainability, and social good, bringing ESG concepts directly to AI solutions development. By bringing together experiences from Brazil and Germany, the session will foster bilateral learning on how to reduce the innovation gap and democratize access to AI technologies.

Session Content and Storyline

The session will follow a coherent storyline from challenges to solutions and impact:

Understanding the SME AI Gap

The session will begin by outlining the structural, technological, and organizational barriers that SMEs face when adopting AI, including technical skills shortages, corporate culture barriers, data availability, cost constraints, difficulty in measuring overcomes, and regulatory/legal uncertainty.

Accessible and Responsible AI Solutions for SMEs

The second part will focus on technological and methodological approaches that lower entry barriers, such as low-code/no-code AI, cloud-based AI services, trustworthy AI frameworks, and human-centered AI design tailored to SME contexts. Also, this part will discuss how AI solutions implementation methodologies could be treated in SMEs in order to guarantee corporate adoption.

Bridging Research, Industry, and Policy

This segment will highlight the role of universities, applied research institutes, and innovation ecosystems in transferring AI knowledge into practice. Public policy instruments, funding schemes, and cross-sector partnerships that incentivize SME digital transformation will be discussed.

Case Studies and Impact Pathways

The session will conclude with concrete case studies from Brazil and Germany, illustrating how SMEs have successfully integrated AI into their operations and value chains, contributing to competitiveness, sustainability, and inclusive economic growth.

Understand to adopt: the importance of making AI explainable

Kevin Bauer, Goethe University Frankfurt

Artificial intelligence (AI) is widely associated with major efficiency gains, improved decision-making, and new opportunities for innovation. Yet these promises are not consistently realized in practice, especially in small and medium-sized enterprises (SMEs). While public and managerial debates often emphasize technical capabilities, this talk adopts an Information Systems perspective and argues that many of the most important barriers to AI adoption in SMEs are socio-technical rather than purely technical. The key issue is not only whether AI can be developed or acquired, but whether it can be integrated into organizational processes and used appropriately by human actors.

This challenge is particularly relevant for SMEs. Compared to larger firms, they often face more limited digital infrastructures and less specialized in-house expertise. At the same time, AI is increasingly becoming a commodity. As access to models and tools expands, value creation depends less on possession of AI itself and more on the ability to embed it effectively into work systems, decision routines, and customer interactions. For SMEs, the innovation gap therefore lies less in access to AI than in its integration and use.

The talk focuses on socio-technical challenges as adoption obstacles. In particular, I will discuss how the black-box nature of AI creates difficulties for appropriate human-AI interaction. In many organizational settings, AI does not fail because outputs are technically unusable, but because human users either do not trust the system, do not understand its boundaries, or rely on it too strongly. Successful adoption therefore depends on calibrated use rather than mere availability. Drawing on my own research, highlight how explainability of AI systems can affect human-AI collaboration, where its strengths lie, and where caution is required.

Human and Data-Driven Intelligence in Technology Roadmapping

Maicon Oliveira, University of Sao Paulo

Technology roadmapping has long been recognised as a collaborative and visual approach for supporting strategic planning, innovation management, and technology decision-making. Traditional roadmapping methods rely heavily on expert interaction, divergent and convergent workshop processes, and value-oriented information processing to connect market needs, technological capabilities, and organisational strategies over time. However, the increasing complexity of innovation ecosystems, combined with the rapid growth of available data and digital technologies, has created new opportunities and challenges for roadmap development and deployment. This talk discusses the evolution of roadmapping toward an integrated human and data-driven intelligence approach. Building on established roadmapping principles — such as agile divergent/convergent processes, collaborative visual tools, and people-centred intelligence — the presentation explores how digital technologies and artificial intelligence (AI) can enhance cognition, creativity, and the generation of strategic insight during roadmapping.

The presentation introduces the concept of hybrid roadmapping, in which human expertise and AI-supported capabilities are combined to improve roadmap preparation, idea generation, information processing, and strategic exploration. Recent advances in large language models (LLMs) are examined in the context of their potential to support creativity-oriented roadmapping tasks through data collection and structuring. Exploratory case studies are presented to demonstrate the practical application of AI-enhanced roadmapping. These include roadmap pre-population and idea generation for connected cars, automotive systems, and manufacturing processes. The studies evaluate prompt-engineering approaches, expert-interaction models, idea quality, feasibility, novelty, and positioning within roadmap structures. Results indicate that AI-assisted roadmapping can significantly increase the number of ideas generated while maintaining high perceived feasibility and supporting novel strategic insights. The findings also highlight the continued importance of expert review and human interpretation in validating and refining AI-generated outputs. In addition, preliminary assessments of LLMs' knowledge of roadmapping concepts are presented using benchmarking questionnaires administered to various commercial and fine-tuned models. The talk concludes by discussing future experiments involving clustering techniques for duplicate-idea detection, extending AI support across complete roadmapping cycles, and developing specialised AI agents for roadmapping applications.

Architecting Human-AI Symbiosis: Designing Virtual Worlds for Collaborative Performance

Alyson Matheus de Carvalho Souza, Federal University of Rio Grande do Norte

Background and Motivation For decades, the relationship between humans and technology in the arts has been primarily utilitarian: we use machines as tools to record, play, or modify our creations. However, as Artificial Intelligence (AI) and Extended Reality (XR) move from the edge into the core of our social and professional lives, a fundamental question arises: How do we move beyond using AI as a tool and begin treating it as a partner? This talk explores a pioneering framework that shifts the focus from "augmented" performance to "integrated" performance. A new paradigm where human agency and artificial intelligence coexist within a shared, navigable virtual world.

The Paradigm Shift: From Product to Process Traditional music and digital art often result in a "fixed" product, be it a recording or a static file. Current research is challenging this by redefining the creative act as an "emergent process." Instead of building a final song, the artist acts as a "World Builder," designing a responsive ecosystem where many movements and decisions by a human participant triggers a unique, real-time response from a generative AI. In this space, the environment itself becomes the instrument, and the act of exploring that environment becomes the performance.

Core Principles of Integrated Performance To bridge the gap between technical innovation and responsible application, this research identifies three critical thresholds for human-AI interaction:

- **Environmental Reciprocity:** Creating a world where physical presence and digital sound are in a constant, mutual dialogue.
- **Embodied Agency:** Moving away from keyboards and clicks to full-body interaction, where a sweep of the arm or a step forward allows the user to "conduct" the digital architecture.
- **Dynamic Emergence:** Ensuring the AI functions as a "co-performer" with its own autonomous creative intent, making every interaction an unrepeatable event.

Democratization and Knowledge Transfer Connecting this to the broader conference theme of "Global Responsible Innovation," this framework offers a path toward democratizing high-level creativity. By building performable worlds rather than complex software, we lower the barrier for non-experts to engage with sophisticated technology. This approach has profound implications for education and research, transforming how we transfer knowledge by allowing learners to physically "perform" within complex systems. **Conclusion** As we stand at the threshold of a new era of generative technology, the goal is not to replace human creativity but to elevate it. By architecting worlds designed for collaboration, we can foster a future where AI and humans don't just work together – they create together.

AI-Enabled Precision Livestock Farming in Integrated Crop-Livestock Systems: Current Applications and Future Challenges at the Gladbacherhof Research

Deise Knob, University of Giessen

The integration of digital technologies and artificial intelligence (AI) into integrated crop-livestock systems (ICLS) represents one of the most significant transformations in modern agriculture, offering new pathways to improve productivity, animal welfare, and environmental sustainability simultaneously. Our objective is to demonstrate how AI-enabled precision livestock farming technologies can be integrated within an ICLS to improve animal health monitoring, nutrient cycling, and greenhouse gas assessment under real farm conditions, based on work conducted at the Gladbacherhof Experimental Farm of Justus Liebig University Giessen (JLU), a long-established ICLS research platform providing the infrastructure, longitudinal datasets, and real-farm conditions required for the development and validation of the technologies presented.

At the Gladbacherhof, automatic milking systems (AMS) continuously record milk yield, milk composition, somatic cell count, and milking frequency, while behavioral sensors monitor rumination time, eating duration, and activity index, providing daily health and welfare information. These data streams feed AI-based algorithms that detect deviations in individual performance and issue real-time health alerts, allowing early identification of mastitis, metabolic disorders, and reproductive problems. Automated feeding platforms optimize ration delivery and monitor feed bunk availability. Sniffer sensors integrated into the AMS also measure individual enteric methane (CH₄) emissions at each milking. In a recent study (Sahraei et al., 2025), LSTM deep learning models predicted individual CH₄ emissions from up to 52 cows, achieving an R² of 0.88 with all features and 0.74 with a reduced, routinely available feature set, indicating their potential for continuous greenhouse gas monitoring.

The ICLS structure also enables closed-loop nutrient management: NIR-equipped slurry spreaders determine nitrogen and dry matter content in real time, generating georeferenced application records combined with soil nutrient maps and GPS-guided tractor data for site-specific management.

In addition, a LoRaWAN-based IoT network connects sensors across the livestock and arable areas, enabling local (edge) AI processing without full dependence on the cloud. Data loggers distributed across the farm generate real-time microclimate (temperature, humidity, precipitation, wind) and soil (temperature, moisture) data, complementing animal-level data for site- and time-specific decisions. This decentralized, low-power architecture reduces latency and reliance on continuous connectivity, an advantage under the variable network conditions typical of rural areas.

Next steps include AI-powered RGB-D cameras for individual DMI estimation, computer vision for body condition score and locomotion scoring, and virtual fencing for precision grazing. Their accuracy under commercial farm conditions, however, still needs validation.

The main challenge remains the integration of these data streams, AMS records, behavioral sensors, CH₄ sniffers, NIR slurry analyzers, GPS data, IoT environmental sensors, and camera-based tools into AI-driven decision-support systems that respond in real time, from sensor acquisition to on-farm decision. Therefore, the Gladbacherhof, as a fully operational ICLS research

farm, represents a suitable platform to demonstrate how digital technologies and AI can strengthen the synergy between livestock and crop production.

Keywords: dairy cattle; automatic milking systems; behavioral sensors; deep learning; computer vision; virtual fencing; NIR slurry analysis; GPS precision agriculture; LoRaWAN; Internet of Things; real-time monitoring

Generative AI for Next-Generation Education, Research, and Knowledge Transfer

Generative AI (GenAI) is reshaping how knowledge is produced, validated, taught, and exchanged. In research, it supports experimentation, data analysis, scientific writing, peer review, and dissemination; in education, it changes how learners and educators create, interpret, and use knowledge; and in knowledge-transfer settings, it mediates exchanges among universities, organizations, public institutions, and communities. These developments can accelerate workflows, reduce linguistic barriers, and widen participation. Yet they also raise questions that technical performance alone cannot answer: what counts as knowledge and creativity? Who retains critical judgment? Whose perspectives become visible? How are benefits, risks, and responsibilities distributed? The session connects education, research, scientific publishing, and universities' third mission - social engagement and societal impact -, with particular attention to scientific workflows, methodological standards, diversity and inequality in global knowledge production, and institutional adaptation. Its common premise is that AI is neither a neutral tool nor an autonomous actor: data, models, interfaces, infrastructures, and uses are embedded in institutional norms, social relations, and structures of power.

Within this session, the speakers' contributions form a deliberate progression from conceptual questions about creativity and knowledge production to applied and normative challenges regarding education and research.

The first contribution draws on Harold Cohen's early work in generative art to interpret the output of GenAI not as a substitute for human creativity, but a catalyst for it. This approach enables us to focus on the question of what "creativity" is in the context of contemporary GenAI and, subsequently, what counts as knowledge, learning, and innovation in an AI-mediated environments. This, in turn, may facilitate knowledge transfer from educational institutions to businesses, and help to produce original, understandable, and assessable outcomes.

The second contribution examines how GenAI is reshaping scientific publishing, peer review, editorial decision-making and knowledge dissemination. While these uses may expand access and efficiency, they concern alongside bias, standardized narratives, delegation of judgment, and global inequalities in epistemic power. Drawing from first-hand experience as an Editor in Chief, the contribution critically reflects issues of these uses affect the scientific publication- and knowledge-dissemination process, including the impact on researchers' analytical independence. It also calls for action in regards to the disclosure and governance of AI use in scientific workflows, putting up methodological standards for evaluating AI-assisted research outputs, safeguarding diversity and relevance in the global knowledge production ecosystem, and interdisciplinary, intersectoral (i.e. academia and businesses), and international collaboration.

Taken together, these first two contributions establish the broad tension between agency, authorship, scientific integrity, and meaningful knowledge-building. Across creative practices, scientific publishing, and education, AI takes on tasks that were once reserved for human judgement, their creativity, and critical evaluation of knowledge. In all circumstances, there is a growing risk of mistaking GenAI output for genuine understanding, efficiency gains for scientific

validity, and the interchangeability of AI-generated output as a substitute for human agency. Read together, the contributions suggest that agency, human creativity, and the potential to learn must be actively renegotiated both within academia, as well as between academia and other societal ecosystems.

The following contributions translate this tension into educational and governance responses, asking which critical capacities, participatory arrangements, and rights-based safeguards are required for responsible AI. This movement—from transformations in creativity and scientific knowledge production to critical AI literacies and rights-by-design—provides the session's central connective thread and structures its Brazilian-German dialogue.

The third contribution develops critical AI literacies for education and sociodigital futures. Rather than reducing literacy to learning with or about AI, this perspective combines theoretical, ethical, practical, and political capacities. It understands AI as a sociotechnical and relational system in which people, algorithms, data, languages, infrastructures, and situated environments interact. This makes anthropomorphism, algorithmic opacity, and distributed human-machine agency central educational concerns. Learners, educators, researchers, and citizens need not only to understand operational principles, but also to assess consequences in context and participate meaningfully in decisions about design, implementation, and governance. From this standpoint, "next-generation" education is not a predetermined technological future, but a contested sociodigital project: which futures are imagined, by whom, for whose benefit, and with whose participation?

The fourth contribution translates these questions into the concrete field of children's rights and AI in basic education. Rights cannot be added as a compliance layer after a system is built; they must inform training data, model design, interaction patterns, transparency, deployment, and evaluation from the outset. Because children and adolescents are learners in ongoing social, emotional, and cognitive development, generic human-in-the-loop governance is insufficient. A children-in-the-loop approach requires prior child-rights impact assessment and co-design with learners, educators, and communities. It also asks whether generated content, recommendations, and performance profiles protect autonomy, dignity, cultural identity, development, and epistemic diversity, or reproduce manipulative patterns and erase local, ancestral, Afro-Brazilian, and other historically marginalized forms of knowledge. This rights-by-design perspective connects applied research, educational practice, and public policy, transforming responsible AI from an optional ethical addition into a design requirement.

The complementary expertise of the session chairs in the sociology of science, higher education research, organizational governance, social psychology, and psychosocial analysis provides a transversal lens connecting individual, relational, institutional, and political levels. From this perspective, GenAI can affect cognition and meaning-making, but also trust, belonging, mental health, psychosocial well-being, recognition, participation, and unequal exposure to harm. Responsible innovation therefore requires transparent and participatory institutional governance, the involvement of affected groups, and evaluation criteria that extend beyond accuracy and productivity to include rights, social impact, and the quality of educational and scientific relationships. The session is open to scientists, policymakers, and practitioners interested in using

AI for education and science while critically examining its social, psychological, institutional, and rights-related implications.

From Output to Agency: What Creativity Can Teach Us About Education in the Age of Generative AI

Caroline Menezes, BSP Business & Law School Berlin

This talk uses the work of the British artist Harold Cohen, one of the earliest pioneers of artificial intelligence in art, as a historical springboard. In the late 1960s, Cohen developed AARON, a rule-based system capable of producing original images. Although grounded in symbolic AI, AARON can be understood as an early generative system. Remarkably, Cohen made a distinction between the production of novel outputs and creativity itself, which he defined as the ability to modify the rules and criteria that shape how a system generates its outputs over time.

This distinction becomes particularly relevant in the context of contemporary generative AI. Today's systems also generate original and convincing outputs, yet they operate without autonomous criteria formation. As a result, they do not engage in creativity in the same sense that human beings do. Revisiting Cohen's understanding of creativity allows us to use it as a guiding lens: if creativity is not simply the production of new outputs, what should count as knowledge, learning, and innovation in an AI-mediated environment? Are we mistaking information reproduction for genuine knowledge building?

Creativity, as a fundamental human capacity (much like curiosity) is central to the learning process. In this context, it can serve as a guiding torch for navigating the expanding use of generative AI in education. Rather than being understood as a replacement for human capabilities, generative AI should be approached as a catalyst. This perspective shifts the focus from "thinking out of the box" as a slogan to the more demanding task of improving knowledge transfer in meaningful ways. For both education and business, the challenge is no longer to produce more results, but to produce extraordinary ones, results that reflect understanding, assessment and originality.

This perspective is developed through case studies conducted at the Institute for Digital Transformation and Innovation at the Business and Law School Berlin, in collaboration with the Mittelstand-Digital Zentrum Zukunftskultur. Our activities explore the use of generative AI in higher education and small and medium-sized enterprises through training programmes, knowledge-transfer formats, evaluation frameworks, and applied case studies. A central principle conducting this work is to place the human aspect at the centre of the debate on digital transformation. This approach is further reflected in the *Digital Transformation and Innovation* book series (Springer Nature), which brings these questions into focus through volumes in Education, Research and AI, such as *Digital Transformation in Knowledge Transfer: Shaping the Future* and *AI and the Ethics of Innovation: Responsibility in Practice*

Who Shapes Knowledge in the Age of Generative AI? Rethinking Peer Review, Publishing, and Global Equity in Next-Generation Research

Felix Holl, Neu-Ulm University of Applied Sciences

Generative artificial intelligence (Gen-AI) is rapidly transforming the foundations of scientific knowledge production. Gen-AI is increasingly embedded in academic publishing—supporting drafting, peer review, editorial decision-making, and dissemination. While these developments promise efficiency gains and broader participation, they also fundamentally reshape how knowledge is produced, validated, and legitimized.

This contribution critically examines the integration of generative AI into scientific publishing and peer review, drawing on first-hand experience as Editor-in-Chief and Associate Editor of two journals as well as research in global health informatics. AI-assisted tools can improve clarity, reduce language barriers, and accelerate editorial processes. At the same time, they introduce risks, including the amplification of existing biases, the standardization of scientific narratives, and the erosion of critical and independent peer review.

Beyond institutional processes, Gen-AI also affects researchers themselves. As AI tools integrate into workflows, critical reflection and intellectual autonomy may be partially delegated to large language models. This shift may affect researchers' analytical independence and the ability of research to generate novel or paradigm-challenging insights.

These challenges are particularly pronounced in global and resource-limited research contexts. It also risks reinforcing existing structural inequalities. Control over data, models, and evaluation standards remains concentrated in high-resource settings, shaping what knowledge is produced and recognized. In this sense, generative AI not only transforms research practices but also redistributes “epistemic power” across global knowledge systems. Importantly, these transformations extend beyond academia. Decision-makers increasingly rely on AI-mediated access to research findings (e.g., via AI agents).

Building on interdisciplinary work in digital public health with international collaborations, this talk argues for an evidence-grounded and equity-oriented approach to integrating Gen-AI into research and publishing ecosystems. It proposes key priorities for action: (1) transparent disclosure and governance of AI use in scientific workflows; (2) methodological standards for evaluating AI-supported research outputs; (3) safeguarding diversity and contextual relevance in global knowledge production; and (4) strengthening interdisciplinary, intersectoral, and international collaboration to co-develop responsible AI practices. Ultimately, the integration of Gen-AI into research and publishing should not be understood solely as a technical innovation, but as a transformation of the social and institutional structures that underpin science. This requires coordinated efforts to ensure AI strengthens the integrity, inclusiveness, and global relevance of scientific knowledge.

Developing Critical AI Literacies (CAIL) for Sociodigital Futures in Education: A Multidimensional Model

Priscila Gonsales, University of Campinas

Given the rapid and often uncritical proliferation of generative artificial intelligence (GenAI) across educational and cultural settings – from schools and universities to museums, research practices, and social institutions – this presentation focuses on the emergent field of critical AI literacies (CAIL) by foregrounding its sociopolitical and critical dimensions. Drawing on the tradition of critical literacy studies, I conceptualise critical AI literacy as an integrated body of theoretical, ethical, and practical knowledge that equips individuals to engage with AI technologies through informed agency. This entails recognising that AI systems are neither neutral tools nor purely technical artefacts, but sociotechnical systems and cognitive assemblages wherein bodies, languages, computational infrastructures, data, and situated environments converge. Accordingly, critical AI literacy aims to provide the conceptual, analytical, and normative resources necessary for individuals to: (a) understand the operational principles and underlying logics of AI systems; (b) assess their social, political, and ethical implications in specific contexts; and (c) participate autonomously and actively in decisions concerning the design, implementation, or integration of these technologies across professional and social spheres.

Yet most literacy and educator training frameworks remain anchored in a reductionist subject-object dualism, treating AI as a passive instrument and the learner as an autonomous rational agent. This prevailing view overlooks critical risks, including anthropomorphism, algorithmic opacity, and the distributed agency shared between humans and machines, that have received insufficient scholarly scrutiny. Such limitations reflect two persistent paradigms within digital literacies: technodeterminism, which frames technology as an exogenous force that simply "impacts" education; and anthropocentrism, which restricts agency to human actors while disregarding non-human participants such as algorithms and data. To move beyond these entrenched positions, I propose an alternative framework grounded in post-humanist and post-phenomenological thought, one that understands CAIL as sociodigital phenomena distributed across cognitive assemblages.

The core problem I address, then, is the inadequacy of existing models, which disproportionately privilege operational and technical competencies (learning "with" and "about" AI) while sidelining the critical formation necessary to interrogate how technology reshapes cognition, social experience, and meaning-making. This gap is further compounded when anthropomorphic metaphors and subject-object dichotomies obscure the relational, opaque, and distributed character of algorithmic agency, perpetuating reductive understandings that undermine critical engagement with these systems. To redress these limitations, I have developed a multidimensional theoretical model for CAIL, grounded in the post-humanist theory of cognitive assemblages (Hayles, 2017; 2024) and the post-phenomenology of technology (Ihde, 2017).

The presentation also introduces the emerging research field of sociodigital futures, which investigates how digital technologies and social life are co-constituted through contested narratives about what is to come. Its main concern is to question which futures are being imagined, by whom, and for whose benefit, actively resisting deterministic and solutionist narratives that exclude

alternative possibilities. From this perspective, citizens and educators must be able to critically intervene in the trajectories of these constructions, defining which sociodigital futures we seek to expand or reject – a task that depends on strengthening Interest-Based Learning Approaches (IBL), which are still underdeveloped. It is about understanding who drives future-oriented narratives and what their implications are in the face of contemporary challenges, such as social inequalities and the climate crisis.

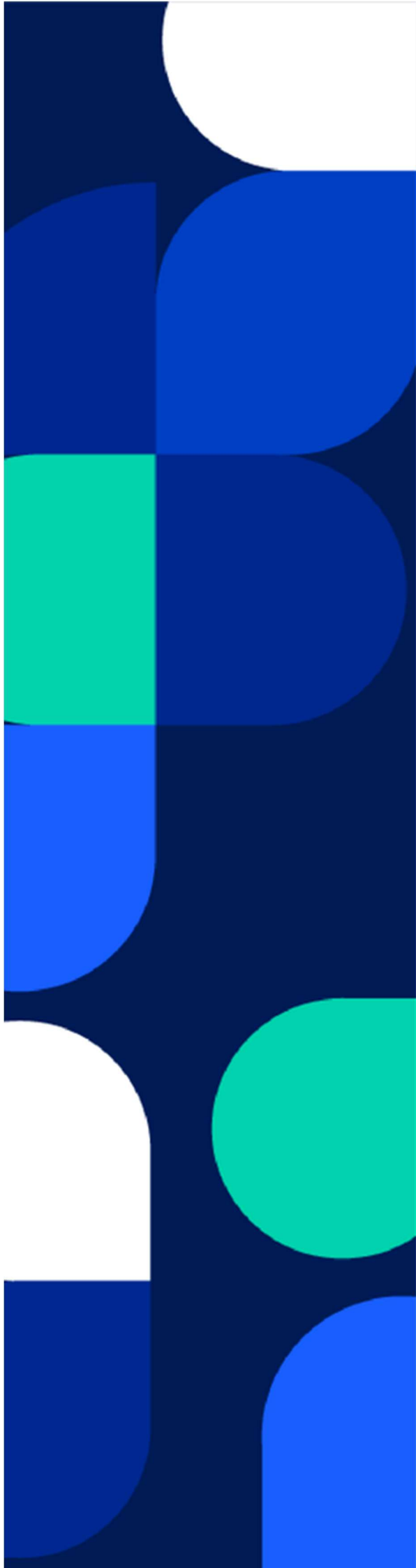
Child Rights by Design in AI applied to Education

Rodrigo Nejm, Alana Institute, Sao Paulo

Rights by design is a concept with radical implications: children's rights cannot be added to a technology after it is built. They must be embedded in the architecture from the beginning, across decisions about training data, model design, interaction patterns, and transparency obligations. Rights by design is not a compliance layer. It is a design commitment that asks, at every stage of development, what a given system does to a child's autonomy, dignity, cultural identity, and development. Rights by design can run as a foundational principle in the context of AI in basic education.

When AI operates in basic education, the challenge becomes concrete in ways that general AI ethics frameworks do not adequately convert into good practices. Children and adolescents as students are special users in social, emotional and cognitive development. This is why the principle of human-in-the-loop, widely used in AI governance, is insufficient here. What the context demands is a “children-in-the-loop” approach, one that treats child rights impact assessment not as a post-deployment evaluation, but as a precondition for any experimentation. The current disruptive moment is also an unrepeatable research opportunity: co-designing the governance and application of AI in schools, with children, educators, and communities, opens a concrete laboratory for applied research on how ethical design can be continuously improved through implementation itself with learning sciences and child development disciplines.

The literature on deceptive and dark design patterns already provides a map of what not to do: manipulative engagement mechanics and interfaces that systematically undermine children's agency. The question now is how to organise a positive agenda around bright and ethical design requirements in educational contexts, and how that implementation process can generate evidence to advance regulatory frameworks in practice, balancing the active promotion of learning benefits with layered protections that minimise harm. A particularly underexplored dimension of this agenda is how AI systems in basic education violate epistemic diversity, erasing local and ancestral knowledge, including Afro-Brazilian culture and history, from generated content, recommendations, and performance profiling. Brazil offers a case study in what balanced regulation can look like and why it matters for innovation. The ECA Digital (Lei 15.211/2025) establishes solidarity liability across the entire digital chain at the same time that the National Education System enables rights-monitored AI experimentation in public schools. The question proposed for collective discussion is how to build a research agenda on children's rights in the design and governance of AI in basic education. Connecting research and public policy, this session invites to build a shared agenda where ethical design is not an option or a “plus” but a requirement and where child rights by design drives responsible AI innovation.



POSTERS

07

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2	Anne Burkhardt	AI & Global (In-)Justice: Research Overview and Outlook
3	Livia Almada Cruz	Time Series Modeling for Digital Phenotyping and Low Back Pain Prediction
4	Washington Cunha	Beyond the "Law of More": Data-Centric Approaches to Sustainable and Responsible Artificial Intelligence in Natural Language Processing
5	Matteo Grosse-Kampmann	Threats and Perceptions of Disinformation in Cognition Altering using AI
6	Marc Hasselwander	From Acceptance to Delegation: Exploring Human-Agent AI Interaction
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10	Dominik Naeher	Backfiring Effects of AI-Driven Information Systems: A Principal-Agent Model with Rationally Inattentive Buyers
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12	Hamza Moluh Njoya	From Fragmented Knowledge to Collaborative Restoration Decisions: Exploring the Potential of Generative AI for Knowledge Integration in Forest Landscape Restoration
13	Elvio Benatto Perino	Microbial production of recombinant bovine caseins for animal-free dairy applications
14	Lara Piccolo	Emancipatory AI: Participatory Approaches to Inclusion and Diversity
15	Livy Real	Aligned for Whom? Toward Culturally Grounded AI Safety Evaluation in Brazil
16	Maximilian Sprang	GenBench - A Safe-by-Design Genomics Foundation Model and Biorisk Benchmark for Generative Sequence Design
17	Susan Stead	From AI Potential to Organizational Value: A Socio-Technical Blueprinting Approach for Designing AI-Enabled Innovation
19	Yao Sun	From Space to Ground: Geospatial Data and AI for Urban and Environmental Challenges
19	Ingrid Winkler	An Agentic AI Framework for Immersive Professional Education: Situated Decisions on the SafeHeight Case
20	Zelun Tony Zhang	Generative AI as Tools for Thought: The Process-Oriented Support Framework

Abstracts of posters

Learning Ecologies Reconfigured by Artificial Intelligence in Higher Education

Alessandra Maieski

The transformations intensified by Digital Culture (Gere, 2008) have fostered new ways of understanding educational processes in Higher Education, since technology is understood as a human, historical, social, and cultural construct. In this context, Learning Ecologies constitute an analytical framework for understanding learning as a dynamic process distributed across different subjects, spaces, resources, and contexts (Barron, 2006; Coll, 2013). The emergence of Artificial Intelligence (AI), especially in its generative form, reconfigures these ecologies by introducing new forms of mediation, production, circulation, and validation of knowledge, while also raising ethical, pedagogical, and epistemological challenges (UNESCO, 2023; Floridi, 2023). This study revisits the findings of a doctoral dissertation (Maieski, 2024), developed within the Graduate Program in Education at the Federal University of Mato Grosso (PPGE/UFMT), supported by a doctoral scholarship from the Coordination for the Improvement of Higher Education Personnel (CAPES), and linked to the Laboratory for Studies on Information and Communication Technologies in Education (LêTECE/CNPq). The original research was conducted at two federal public universities in the state of Mato Grosso, Brazil, adopting a qualitative approach (Denzin & Lincoln, 2000) and Virtual Ethnography (Hine, 2007), involving institutional managers, program coordinators, professors, and students. Although the original investigation focused on the constitution of Learning Ecologies in Higher Education, this study reinterprets its findings in light of the contemporary transformations associated with Artificial Intelligence, enabling a reinterpretation of the three structuring pillars identified in the research: (i) social interaction and pedagogical mediation; (ii) circulation of materials and knowledge construction; and (iii) (re)constructed lessons. The findings indicate that AI operates across these three pillars, reconfiguring the forms of interaction between professors and students, the processes of knowledge production and circulation, and the dynamics of learning. At the same time, they highlight the need to strengthen principles related to ethics, transparency, authorship, scientific integrity, and the development of critical thinking, ensuring that these technologies are integrated through pedagogical intentionality and responsible educational practices (UNESCO, 2023; Floridi, 2023; Brazil, 2026). It is concluded that Artificial Intelligence is not an external element to Learning Ecologies but rather becomes part of their very constitution, reconfiguring the relationships among subjects, technologies, resources, and educational practices. From this perspective, Learning Ecologies emerge as a consistent theoretical framework for understanding the contemporary transformations of Higher Education and for supporting pedagogical practices developed with pedagogical intentionality, grounded in ethical, critical, and innovative principles.

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AI & Global (In-)Justice: Research Overview and Outlook

Anne Burkhardt

Artificial Intelligence (AI) has long been recognized as a highly transformative technology. At the same time, critical and decolonial research has shown that the benefits and costs of AI are not distributed equally across the globe, nor is the opportunity to actively shape the development of the technology and related debates. In short, the Norths benefit disproportionately from the current development and deployment of AI, e.g. through advanced connectivity, robust digital infrastructures, widespread technological access, and the economic gains generated by Northern-based Big Tech companies. By contrast, the Souths bear a disproportionate share of the associated social, economic, and environmental costs, including the extraction of raw materials for digital hardware, the exploitation of labor in the clickwork sector, practices of data colonialism, the disposal of digital waste, and the environmental damage caused by the high energy consumption and CO₂ emissions of AI data centers. Beyond these material inequalities, there are also far-reaching epistemic injustices. On the one hand, these injustices concern the unequal visibility and recognition of knowledge and experiences from the Souths related to AI across scientific, ethical, and fictional discourses. On the other hand, injustices manifest in the marginalization of underrepresented languages, value systems, and forms of knowledge through data-driven decision-making systems and generative AI technologies such as image generators and large language models. To adequately address the global challenges arising from the AI-driven transformation of contemporary societies, it is essential to incorporate diverse perspectives rooted in different regional, cultural, and epistemic contexts.

The postdoctoral research project “AI & Global (In-)Justice” explores how structural injustices—including infrastructural, environmental, epistemic, narrative, and engagement injustices—manifest and are reinforced in the context of AI development and use as well as related (science) communication, and how these injustices can be addressed. To this end, the project foregrounds perspectives, concepts, and approaches from “the Souths”—that is, from individuals or groups that “suffer discrimination, and/or enact resistance to injustice and oppression” (Milan & Treré 2019, 325)—while fostering equitable exchange, collaboration, and mutual learning with researchers across the Souths and Norths.

In the first phase, the project developed a theoretical framework for understanding dimensions of global (in)justice in the context of AI and conducted empirical case studies focusing on specific dimensions of justice. The first study examines how AI technologies perpetuate and reinforce colonial power asymmetries as well as racialized and gendered forms of discrimination, drawing primarily on decolonial scholarship and activism (Burkhardt 2023). The second study foregrounds the material infrastructures on which AI depends and of which it has become a part, and analyzes their implications for global ordering from a global justice perspective (Burkhardt 2025a). The third study investigates how AI is imagined and represented in Latin American science fiction films (Burkhardt 2026) and explores, through interviews with filmmakers, how context-driven experiences have shaped their perspectives on AI and how structural conditions have rendered their work largely invisible (Burkhardt 2025b).

The second phase of the project is dedicated to exploring intercultural perspectives on science communication about and with AI, as well as to fostering critical AI literacy and societal resilience in the face

of persuasive technologies such as LLM-based chatbots. In this context, international collaboration, intercultural exchange, and mutual learning among partners from the Norths and Souths are essential. This commitment is reflected in the establishment of an interdisciplinary, international working group on AI and Global Justice and in an ongoing grant application on AI and persuasion involving collaborators from both regions.

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Knowledge, unlocked in every language Christian Leiterer

Written knowledge is locked behind two barriers: a language the reader cannot read, and a scan the machine cannot read. This poster presents two open, AI-assisted tools that address both from a single shared pipeline. Tool A transcribes or translates any PDF into any of 100+ languages across seven scripts, preserving headings, tables and columns. Pages are cleaned, batched and passed to a vision LLM in one streamed call; a page-break protocol and rolling context rejoin sentences and tables that span pages. Export fonts are chosen by scanning the Unicode code points of the output rather than the language label, so CJK, Arabic, Devanagari, Cyrillic, Greek, Hebrew and Thai each receive the correct face, with right-to-left layout where needed. Tool B lowers the word- and character-error rate of ALTO/hOCR from scanned books. Only words below a confidence threshold are revised: the model re-reads the cropped image region and returns word-ID to text, so every bounding box and the original geometry survive untouched. The model is instructed to transcribe, never edit, which suppresses hallucinated corrections. Results are measured with standard and cleaned WER/CER via jiwer, logged per stage, and exportable as ground truth. Both tools sit behind one interface over Gemini, GPT, Claude or fully offline Ollama, so sensitive or rare material need never leave the institution. The approach is deliberately hybrid, open and sustainable: cleaner OCR turns scanned archives into citable, full-text-searchable research data, and any reader reaches a document written in any other language, free of charge and auditable at every step.

Backfiring Effects of AI-Driven Information Systems: A Principal-Agent Model with Rationally Inattentive Buyers

Dominik Naeher

Digital markets are increasingly shaped by AI-based information technologies, such as chatbots, recommender systems, and filtering algorithms, that curate and prioritize information for users. While these tools can help improve decision-making, a growing body of evidence shows that more or better-targeted information can sometimes reduce consumer welfare. This paper provides a novel theoretical framework to understand when and why such backfiring effects arise.

We develop a model of a multi-category seller interacting with consumers who face limited attention and must allocate a fixed capacity to process information about uncertain product match values. Attention devoted to one category improves decisions there but crowds out attention elsewhere, generating cross-category effects that are difficult to predict a priori but that are key to understanding the welfare impact of information provision policies. These trade-offs are particularly relevant in AI-mediated environments, where algorithms actively steer user attention across products, categories, and platforms.

Our analysis identifies a potential deadweight loss and highlights conditions under which policy intervention is warranted. We distinguish two types of instruments: (i) instruments that reduce the cost of learning (intensive margin of uncertainty reduction), such as simplifying information via AI tools, and (ii) instruments that increase the share of fully informed consumers (extensive margin), such as unbiased recommender systems and filtering tools. Strikingly, we show that the former often backfire, while the latter tend to improve consumer welfare.

From a managerial perspective, we show that the amount of attention allocated to a product category can be non-monotonic in the information processing cost in that product category, and that the desired outcomes in a specific category can be achieved indirectly through information provision in other categories. Our findings are of a general nature and offer new insights for the design and regulation of AI-driven information systems in strategic settings across a broad set of contexts including business, trade, and education. They emphasize that optimizing local information provision, such as improving recommendations in a single category, can have unintended negative effects across the broader choice environment.

Microbial production of recombinant bovine caseins for animal-free dairy applications

Elvio Benatto Perino

The increasing demand for sustainable protein sources has intensified interest in cellular agriculture and microbial production platforms for animal-free food ingredients. Caseins represent approximately 80% of the total protein content of bovine milk and are essential for the functional properties of dairy products, particularly cheese texture, micelle formation, and nutritional value. However, recombinant casein production remains challenging due to protein aggregation, post-translational modifications, host-dependent proteolysis, and downstream processing requirements.

In this study, microbial systems based on *Bacillus subtilis* and *Escherichia coli* were evaluated for the recombinant production of bovine caseins. *B. subtilis* was investigated as a GRAS production host with secretion potential and suitability for high-cell-density cultivation in defined mineral salt medium. Fed-batch cultivation in a 42 L bioreactor enabled production of His-tagged α S1-casein, reaching a maximum titer of 56.9 mg/L and a product-to-biomass yield of 2.16 mg/gCDW. Downstream processing using immobilized metal affinity chromatography resulted in partial purification of the recombinant protein. In parallel, *E. coli*

BL21-Gold (DE3) was used for intracellular β -casein production. Initial expression in the parental strain did not result in detectable casein, suggesting degradation by intracellular proteases. Deletion of the protease genes *clpP* and *clpQ* enabled successful β -casein production. Fed-batch cultivation in a 17 L bioreactor reached a maximum titer of 1.2 g/L and a product-to-biomass yield of 50.85 mg/gCDW. These results demonstrate the potential of bacterial cell factories for recombinant casein production and highlight host engineering, fermentation optimization, and downstream processing as key steps toward scalable animal-free dairy protein production.

Computational Sociolinguistics for Data and Model Analysis

Filip Miletic

This poster presents the general context for my research, which takes as its starting point the observation that different people use language in different ways; importantly, these differences in language use can be systematically explained by sociolinguistic factors such as speakers' sociodemographic properties, their patterns of interaction, and the social meaning conveyed by their linguistic choices. From one angle, my research develops computational methods and data resources to scale up empirical sociolinguistic research; from another, it draws on sociolinguistic insights to critically investigate the performance of language technologies such as LLMs. This poster presents two ongoing studies which respectively illustrate these lines of work.

The first study addresses a recurrent methodological issue in computational analyses of sociolinguistic phenomena: how can we obtain large-scale collections of human-produced data which also include rich text-level sociolinguistic metadata? The present study draws on the sociolinguistic principle of audience design, i.e., the notion that speakers adapt their language use not only as a function of their own characteristics but also those of their audience. As a starting point, we leverage an existing large-scale corpus of radio transcripts from the United States. Rather than attempting to infer sociolinguistic information on the recorded speakers – which raises ethical concerns due to their unawareness of this step – we enrich the original corpus with data on the population to which the radio stations are broadcasting, obtained from the United States Census as a highly reliable and publicly available source of information. This enables us to create subsets of the original corpus stratified for sociodemographic properties of the broadcast areas (e.g., those with high vs. low degree of education as compared to the overall distribution). We then systematically evaluate whether the obtained audience-level patterns align with previously established speaker-level sociolinguistic behaviors; and we further explore novel insights that this dataset may provide.

The second study draws on established sociolinguistic knowledge to investigate the robustness of LLMs in an educational scenario, specifically examining models' ability to provide metalinguistic information (analysis and explanation of linguistic patterns). Shifting away from the prevalent focus on English-language data in prior work, we focus on the more challenging linguistic context of the BCMS (Bosnian, Croatian, Montenegrin, Serbian) linguistic area. This context is characterized by distinct national language varieties which are mutually intelligible but subject to different prescriptive norms (institutionally established understanding of correct and incorrect linguistic behaviors). We propose a comprehensive experimental setup to analyze metalinguistic knowledge on multiple levels of linguistic structure (phonology, morphology, morphosyntax, syntax, and lexicon), systematically examined across five tasks of increasing difficulty, and further controlling for instances where the national prescriptive norms differ vs. align. This work will overall provide important insights into LLM performance in a realistic educational use case scenario, and more generally highlight broader implications for model robustness in complex sociolinguistic contexts.

From Fragmented Knowledge to Collaborative Restoration Decisions: Exploring the Potential of Generative AI for Knowledge Integration in Forest Landscape Restoration

Hamza Moluh Njoya

Forest Landscape Restoration (FLR) requires the integration of knowledge from diverse stakeholders, including farmers, local communities, researchers, development practitioners, and policymakers. However, restoration decisions are often challenged by fragmented knowledge distributed across scientific, local, institutional, and policy domains, differing stakeholder priorities, and limited mechanisms for knowledge integration and exchange. At the same time, generative Artificial Intelligence (AI) is rapidly transforming how knowledge is synthesised, communicated, and applied across sectors. This poster explores how generative AI could support knowledge integration in FLR by bringing together scientific evidence, local knowledge, policy information, and stakeholder perspectives. Generative AI has the potential to support this integration by synthesising information from scientific literature, policy documents, project reports, and stakeholder inputs into accessible, context-sensitive knowledge that can inform restoration planning and collaborative decision-making. Building on experiences from forest landscape restoration and food-security research in West Africa, the poster examines the opportunities and challenges of AI-assisted knowledge integration in smallholder landscapes. It addresses three key questions: (1) What knowledge and information needs exist among different FLR stakeholders? (2) Which AI tools are already being used and for what purposes? (3) How can future AI-enabled approaches better support collaborative restoration planning and decision-making? The objective is to stimulate discussion on the responsible use of AI for stakeholder engagement, knowledge co-creation, and context-specific restoration planning, while identifying opportunities for interdisciplinary collaboration and future AI-enabled decision-support systems for FLR.

An Agentic AI Framework for Immersive Professional Education: Situated Decisions on the SafeHeight Case

Ingrid Winkler

Professional education in safety-critical domains faces a persistent knowing–doing gap. Workers may understand regulations and answer procedural questions, yet still struggle to recognize hazards, interpret changing conditions, resist productivity pressure, and decide when an activity must be interrupted. Digital courses, chatbots, quizzes, and immersive simulations can support this process, but they are usually implemented as disconnected resources. Evidence generated in one activity rarely modifies the next, while assessment often privileges recall or user satisfaction rather than situated decision quality. The challenge is how to transform heterogeneous educational technologies into a coherent, adaptive, and evidence-driven pathway that develops autonomous professional judgment.

Our previous research provides complementary foundations for this challenge. A review of immersive technologies for industrial safety training identified opportunities for realistic, repeatable, and risk-free practice, alongside challenges involving evaluation validity, user factors, and transfer to real work [1]. Preliminary design guidelines emphasized alignment among learning objectives, task conditions, performance evidence, and evaluation instruments [2]. Related studies reinforce the relevance of systematic VR usability evaluation [3], visually structured and gamified communication for professional audiences [4], and the future incorporation of non-contact physiological evidence through machine-learning-based heart-rate estimation [5].

Building on these foundations, we propose an evidence-driven Agentic AI framework for adaptive immersive professional education. A goal-oriented educational agent coordinates modular instructional materials, a

source-grounded chatbot, an expert interview agent, adaptive assessment, immersive task rehearsal, AI-assisted debriefing, and a final performance certificate. Each stage produces structured evidence about the learner's knowledge, misconceptions, readiness, decisions, and performance. These outputs update a learner model and determine the entry level, difficulty, prompts, feedback, remediation, or progression criteria used in the next stage. Adaptation is bounded by a curated knowledge base and expert-defined assessment rules. The contribution is therefore an architecture that connects instruction, assessment, experience, and reflection through traceable evidence.

Competent work-at-height performance requires more than memorizing regulatory provisions: learners must inspect conditions, recognize hazards, evaluate protective measures, respond to environmental changes, and make stop-or-continue decisions under production pressure.

The framework is instantiated in SafeHeight Agentic Tutor, a prototype for work-at-height professional education that includes an AI-generated e-book and short videos introducing a curated NR-35 knowledge base, and a source-grounded chatbot answering learner questions. Interaction evidence supports an initial classification as basic, intermediate, or expert, which configures an AI-mediated interview and adaptive quiz. Reports from these stages inform the difficulty, prompts, and decision conditions of an immersive VR scenario. Learners identify hazards and decide whether the task may proceed, requires corrective action, or must stop. Their performance then feeds an AI-assisted debrief that relates observed actions to regulatory principles, identifies misconceptions, recommends targeted review, and supports issuance of a final performance certificate.

Our work presents the reference architecture, evidence flow, and current prototype. The framework can be replicated by replacing the knowledge base, agent prompts, scenario, performance criteria, and debriefing logic. Potential domains include construction safety, industrial maintenance, confined spaces, emergency response, and healthcare procedures. Future evaluation will examine learning gains, decision quality, transfer, usability, trust, explainability, and multimodal evidence. The expected contribution is a reusable model for transforming fragmented AI and immersive resources into an adaptive, human-centered, and evidence-based professional learning pathway.

Accelerating AI Adoption in Agri-Food SMEs: The Role of Shared Agricultural Data Infrastructure

John Mulwa Ndisya

The need for resilient and sustainable food supply chains has accelerated digital transformation, making artificial intelligence central to innovation in agriculture. Small and medium-sized enterprises (SMEs) are vital to global economies and food security, yet face significant barriers to AI adoption. These include high upfront investment costs, limited access to high-quality data, insufficient technical capacity and complex regulatory requirements. This poster argues that shared, AI-ready agricultural data infrastructure is essential for SMEs to adopt and integrate AI, supporting digital transformation across agri-food value chains. The central argument is that wider AI adoption depends on the availability of shared, AI-ready agricultural data suitable for specific analytical or operational purposes, relevant to local contexts, linked to quality-assured reference measurements, in machine-readable formats and with documented metadata, provenance and intended use cases. Data sharing should also be governed by explicit technical, legal and organisational frameworks, such as the General Data Protection Regulation (GDPR) for privacy compliance, data trusts and data-sharing agreements for managing usage rights, data standards such as ISO 19115 and Darwin Core for metadata and FAIR Data Principles to ensure interoperability and transparency.

Agri-food SMEs operate throughout the value chain, from agricultural inputs and production to processing, distribution, and related services. Data in these value chains is often distributed across public and private repositories in structured and unstructured formats, including images, statistics, paper records, sensor streams, and models. This fragmentation limits access, integration, and reuse. Data-driven technologies and services are vital in agriculture for production and supply chain planning, credit assessments and regulatory compliance. AI technologies and services rely on aggregated, high-quality datasets at a scale and cost that individual SMEs cannot achieve on their own. Therefore, data sharing provides tangible benefits to SMEs and returns value to data providers, including farmers at the very basic level.

Under the European Partnership Agriculture of Data (AgData), we are harnessing agricultural production, Earth observation, environmental, policy, administrative and digital technology data to support innovation and transformation across the European agri-food sector. Key data sources include farm records, Farm Management Information Systems, satellite imagery, public statistics, sensors, Internet of Things devices, and weather, climate, soil, water, biodiversity and land-use information. The Leibniz Institute for Agricultural Engineering and Bioeconomy provides AgData with use cases involving computer vision, machine learning, digital twins, and reference datasets. Examples include annotated plant and weed imagery for agronomic and biodiversity monitoring, sensor and emission data for dairy housing systems, and digital twins for climate control and energy management in post-harvest storage systems and logistics.

The work under AgData is a practical demonstration of how shared agricultural data infrastructure can provide a foundation for AI services that SMEs can tailor to their operational and commercial needs. However, addressing the challenges of data standardisation across heterogeneous sources, achieving stakeholder buy-in, addressing concerns about data privacy, establishing trust among participants, aligning incentives, and ensuring fair sharing of benefits are essential for realising the full benefits of shared agricultural data for SMEs.

Emancipatory AI: Participatory Approaches to Inclusion and Diversity

Lara Piccolo

As Artificial Intelligence (AI) permeates processes, decision-making and education across all domains of life, communities already marginalised in benefiting from digital technology face a widening gap. Despite growing consensus that users and stakeholders should participate more in AI design so that systems better reflect their values, preferences, and needs, recent HCI and participatory design scholarship highlights significant variation in how users' agency and power are negotiated within AI projects. Two risks in particular stand out: tokenism, which is including marginalised groups just enough to appear inclusive without giving them real influence over outcomes, and cognitive imperialism, which is the imposition of dominant ways of knowing on communities whose knowledge and experience differ.

Addressing these gaps, this research advances the concept of emancipatory AI: conceiving, designing, building and governing systems through participatory processes that prioritise inclusion and diversity. The work draws on two methodological foundations: Socially Aware Design (Baranauskas, 2014) and Participatory Design as described by Bødker et al. (2022). In both traditions, the inclusion of diverse people and perspectives underpins the creation of digital technologies, and the design process is understood to extend beyond technical concerns to encompass both formal aspects, such as rules and regulations, and informal ones, such as people's values, beliefs, and other cultural traits. Together, these frameworks support a methodological reframing that treats the inclusion of individuals with diverse needs not as a matter of technical accessibility but as a fully participatory and socially aware challenge.

We propose two case studies, in Germany and Brazil, to advance methods and tools for participatory design and appropriation of AI-based tools:

The first is grounded in the project INLAB - the Inclusive Future Lab for Work, Profession and Technology - funded by the German Ministry of Labour and Social Affairs. INLAB targets people with disabilities transitioning from sheltered vocational training environments into the mainstream labour market and aims to develop a replicable, open-access toolbox of inclusive methods and tools to facilitate the inclusion of employees with diverse needs in organisations.

The second examines AI in informal education in native languages across Brazilian indigenous territories, where access to technology and data sovereignty are kept at the centre.

By foregrounding cultural specificity, sustained collaboration, and community accountability, this work charts a path for technology designers committed to AI that not only avoids harm but actively advances social justice. Together, the two case studies share similar challenges of AI literacy and technology appropriation, demonstrating that emancipatory AI is a situated practice that requires continuous negotiation with, and accountability to, the communities it serves.

Time Series Modeling for Digital Phenotyping and Low Back Pain Prediction

Livia Almada Cruz

Time-series-based biomedical data modeling has great potential for healthcare applications. In people with musculoskeletal conditions, such as low back pain, one of the leading causes of disability worldwide, physical activity is a relevant factor in symptom prevention and the reduction of disability. Time-series analysis combined with digital health tools can contribute to understanding the relationship between different physical activity profiles, lifestyles, and the occurrence of pain and disability, enabling the identification of behavioral phenotypes and factors associated with health outcomes. Wearable devices and smartphones enable continuous data monitoring, providing information on physical activity, behavior, and lifestyle under real-world conditions. This work is part of the research line on Anomaly Detection in Biomedical Data within the AI4Wellness Project, a partnership between the Federal University of Ceará and Samsung Brazil. The research proposes developing and evaluating computational models for human activity recognition, identification of physical activity and lifestyle phenotypes, and prediction of low back pain flare-ups based on longitudinal data collected from wearable devices. This poster presents the main results of this research, highlighting the computational approaches developed.

Aligned for Whom? Toward Culturally Grounded AI Safety Evaluation in Brazil

Livy Real

Frontier AI models are developed, aligned, and evaluated within a limited set of cultural and linguistic contexts, yet are increasingly deployed worldwide. We argue that this creates a structural limitation operating at three levels: technical, because safety evaluations degrade outside English; theoretical, because prevailing alignment frameworks assume a universal notion of human values; and governance, because model safety remains largely determined by private laboratories. Using the Brazilian context as a motivating case, we propose a research agenda for culturally grounded AI safety consisting of three complementary directions: participatory dangerous capability benchmarks, a nationally accountable evaluation infrastructure, and adaptive evaluation methodologies capable of evolving alongside frontier models. Rather than proposing

a universal notion of AI safety, we advocate a procedural framework in which affected populations participate in defining and evaluating the risks relevant to their own social and institutional contexts.

From Acceptance to Delegation: Exploring Human–Agentic AI Interaction

Marc Hasselwander

Agentic AI marks a significant shift in human–AI interaction, moving beyond reactive chatbot-based systems toward autonomous, goal-directed agents capable of multi-task execution with minimal human intervention. As these systems become increasingly embedded in everyday life, understanding how users perceive, accept, and delegate tasks to agentic AI becomes essential. Drawing on survey data from 407 experienced AI users in the Philippines, this combined study examines human–agentic AI interaction through two complementary perspectives: the acceptance of deeply personalized AI super assistants and the anticipated delegation of everyday and work-related tasks to autonomous AI agents.

The first study investigates user acceptance of AI super assistants through an adapted AIDUA-based model that incorporates perceived risk. Using PLS-SEM and necessary condition analysis (NCA), the study identifies key drivers of acceptance, including performance expectancy, effort expectancy, hedonic motivation, perceived novelty, and social influence, while also highlighting the role of perceived risk in shaping user evaluations. These findings suggest that users do not assess agentic AI merely in terms of functional utility, but also through experiential, emotional, and trust-related considerations.

The second study extends this perspective by examining how users imagine delegating tasks to agentic AI systems. Using latent class analysis across 12 everyday and work-related activities, the study identifies heterogeneous delegation profiles, including Creative Task Delegators (36.4%), Everyday Convenience Seekers (6.6%), Socially Oriented AI Agent Users (19.2%), and Web-Browsing Automators (37.9%). These segments reveal that willingness to delegate is not uniform, but varies according to users' orientations toward autonomy, cognitive offloading, social interaction, and perceived usefulness. The membership model further indicates that age, socioeconomic status, social influence, and hedonic motivation shape delegation preferences, suggesting that both demographic and motivational factors influence how users envision future interactions with autonomous AI systems.

Together, the two studies provide an integrated view of the emerging transition from agentic AI acceptance to delegation. While acceptance research explains why users may adopt deeply personalized AI assistants, delegation research reveals what users may allow these systems to do on their behalf. This combined perspective contributes to ongoing discussions on responsible AI design, human oversight, trust calibration, and the redistribution of agency between humans and autonomous systems. By situating agentic AI within the Philippine context, the studies also offer early empirical evidence from a digitally engaged emerging market, highlighting the cultural and social dimensions of human–agentic AI interaction.

Threats and Perceptions of Disinformation in Cognition Altering using AI

Matteo Grosse-Kampmann

Generative AI and Brain Computer Interfaces (BCI) are more intertwined each day with modern wearable BCIs already integrating generative AI, for example to analyze emotional or affective states. While useful scenarios like motor-control or safety enhanced scenarios are possible using these devices, this development leads to several threats to the privacy and security of human cognition. These intertwinings allow potential adversaries to access human as well as artificial cognition. This threat is acknowledged as a means of

modern war by NATO or other military as well as political organs already. Especially in the context of disinformation this becomes especially concerning as disinformation can enter cognition and behavior on various levels from the societal over the group to the individual level due to the ubiquitous interconnectedness of humans and artificial intelligences. Potential adversaries might target human or artificial cognition as they can scale and individualize their tactics using artificial intelligence. There is currently no threat model available for a system that combines these three technologies in regards to disinformation. I would like to develop a model that integrates the three technologies and models threats using for example the STRIDE method and evaluate such a model using cross-cultural insights from Brazil and Germany and compare them. Furthermore, the project should develop certain countermeasures to address the identified threats.

GenBench - A Safe-by-Design Genomics Foundation Model and Biorisk Benchmark for Generative Sequence Design

Maximilian Sprang

Generative models of biological sequence have crossed a threshold. In 2025, King and colleagues used the genome language model Evo to design bacteriophages whose genomes yielded viable, infectious progeny: whole-genome design, executed by a model. The capability is real; the safety infrastructure is not. A 2026 Epoch AI survey of 1,196 biological AI models found that only 3.2% carry any documented safeguards and only 2.5% underwent any pre-release risk assessment, against roughly 95% for frontier language models. Biology and Health are 'high-risk domains', but in the AI field there are almost no established safeguards.

We propose GenBench, a reference genomics foundation model (GFM) built safe-by-design and coupled to a reproducible biorisk benchmark and a tiered-access governance platform. It has four elements. First, a GFM spanning DNA, RNA and protein, with biological priors and biosecurity constraints embedded in pre-training and in the architecture rather than applied as an output filter. Second, an explainable-AI interrogation interface that scores risk and exposes mechanism, supporting counterfactual analysis of how specific sequence changes alter hazard. Third, a reproducible biorisk benchmark measuring viral host shift and humanisation potential, the generation of new-to-nature toxic components that evade BLAST-based screening at synthesis providers, and One-Health impacts across human, animal and agricultural contexts. Fourth, a tiered-access platform through which regulators, synthesis vendors and researchers can use the model under risk-proportionate controls.

The consortium, so far, spans nine institutions across four European countries, with public-health and high-containment partners including the Robert Koch Institute. At BRAGFOST I would especially like to discuss pandemic surveillance and One-Health risk with Brazilian colleagues. Spillover-class hazards are only visible at that level, and a benchmark built without Southern-hemisphere pathogen and agricultural context will not become a global reference.

Strategic AI Adoption in Construction: Bridging Technology, People, and Organizational Change

Niels Bartels

The successful implementation of Artificial Intelligence (AI) in construction requires more than technological innovation as it demands organizational transformation. This research explores how construction companies can effectively integrate AI by combining empirical evidence with established change management principles.

Based on survey data from construction professionals and expert interviews, the study identifies the most influential drivers and barriers to AI adoption and develops an industry-specific implementation framework.

The proposed model emphasizes human-centered AI adoption through communication, stakeholder participation, continuous learning, and iterative implementation. The results contribute practical guidance for researchers and practitioners aiming to accelerate digital transformation in the built environment and provide a foundation for future international collaborations on AI, digital twins, lifecycle data management, and sustainable construction.

Beyond the construction sector, the proposed framework provides a transferable approach for AI implementation in other project-based and highly regulated industries, such as manufacturing, healthcare, energy, and public infrastructure. By integrating technological readiness with organizational change management, the framework addresses common challenges including workforce acceptance, governance, skills development, and cross-disciplinary collaboration. The findings contribute to a broader understanding of human-centered AI adoption and support organizations in translating AI innovations into sustainable organizational transformation.

From AI Potential to Organizational Value: A Socio-Technical Blueprinting Approach for Designing AI-Enabled Innovation

Susan Stead

Artificial intelligence (AI) offers substantial opportunities to enhance organizational processes and create new forms of value. However, many AI initiatives struggle to move beyond technological feasibility because they insufficiently consider the organizational contexts, workflows, and human actors in which these systems are embedded. This research addresses this challenge by developing a socio-technical blueprinting approach for designing and prototyping AI-enabled innovation.

Building on service blueprinting and design science research, the approach enables organizations to systematically map actors, processes, frontstage and backstage activities, and the intended tasks and roles of AI systems. By explicitly representing AI as an actor within organizational processes, the framework supports the identification of meaningful use cases, clarifies responsibilities, and facilitates alignment between technological possibilities and organizational needs.

The approach is developed and refined through a four-year interdisciplinary AI project in polytrauma care and applied across multiple hospital processes, including documentation, communication, and patient support. Findings demonstrate how blueprinting can make invisible work, coordination challenges, and potential AI contributions visible while supporting dialogue among technology developers, domain experts, and organizational stakeholders.

The research contributes to responsible AI innovation by highlighting that successful AI adoption requires more than advanced technologies. Instead, organizations need approaches that bridge the gap between AI capabilities and socio-technical realities, enabling trustworthy, human-centered, and sustainable AI transformation.

Beyond the "Law of More": Data-Centric Approaches to Sustainable and Responsible Artificial Intelligence in Natural Language Processing

Washington Cunha

This poster presents a viable, sustainable scientific alternative to the dominant "Law of More" paradigm in Natural Language Processing (NLP), which entails alarming financial, hardware, and ecological costs, such as the 552 tons of CO2 emitted during the pre-training phase of GPT-like approaches. Aligned with the Responsible and Green AI guidelines of INCT-TILDIAR, the research focuses on data-centric AI approaches, utilizing advanced data engineering techniques like instance selection and active learning, to intelligently curate training data by eliminating noise and redundancy. Experimental results using the innovative E2SC and biO-IS algorithms demonstrate that it is possible to reduce training data volume by up to 60% without compromising language model effectiveness, enabling training processes up to 6.1 times faster and paving the way for more efficient, accessible, and potentially more sovereign AI development in Brazil and beyond.

From Space to Ground: Geospatial Data and AI for Urban and Environmental Challenges

Yao Sun

Urban and environmental challenges require information across scales: from satellite-based Earth observation to ground-level imagery, maps, text, and local knowledge. Remote sensing provides repeated, large-scale observations of land cover, urban growth, infrastructure, environmental exposure, and disaster impacts. However, many decision-relevant questions cannot be answered from satellite data alone, especially when they involve building-level exposure, informal settlements, local vulnerability, or the interpretation of complex urban environments. This poster presents our work on geospatial data and AI for urban and environmental understanding. Recent research includes vision-language models for extracting building exposure and risk-related information, synthetic datasets for building damage detection, and simulated urban data generation for object-level urban reconstruction and geospatial AI evaluation. These researches connect remote sensing, street-level imagery, synthetic data, and multimodal AI to support more scalable and interpretable analysis of cities and disaster-prone environments. We are particularly interested in Brazil-Germany collaborations on multimodal urban and environmental datasets, informal settlements such as Brazilian favelas, disaster-risk assessment, synthetic data for rare or hard-to-label scenarios, and trustworthy AI for public-sector decision support.

Generative AI as Tools for Thought: The Process-Oriented Support Framework

Zelun Tony Zhang

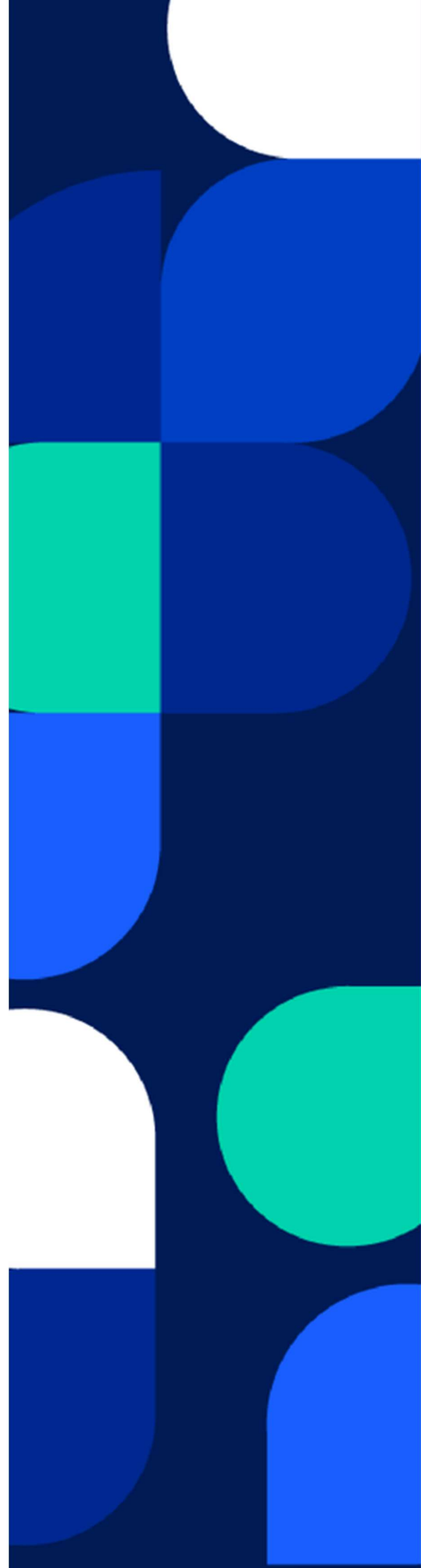
Generative AI (GenAI) is expected to automate more and more knowledge work, resulting in widespread fear of massive job losses and deskilling. In education, a similar common concern is that GenAI will undermine learning as students can offload most coursework to AI. The underlying question with profound economic, societal, but also philosophical implications is: What is left for humans when AI does all the cognitive work? At the same time, the current state of AI adoption is still far off from one where AI takes over all cognitive work from humans. Many, if not most GenAI projects fail to make a significant economic impact. Low-quality GenAI outputs can even harm productivity, for instance when they are passed on to coworkers without substantial editing.

Both the concerns about replacement of human cognition and the challenges in adopting AI productively are consequences of the same root cause: GenAI is often designed and used to automate tasks as much as possible, which makes it hard for humans to contribute their own thinking. However, automation is not an

inevitable result of technological progress, but a choice by AI developers, deployers, and/or users. While full automation can be a good choice in some situations, in many other contexts it is not. Thus, my research is motivated by the following question: How can we build GenAI tools that do not undermine human cognition through full automation, but rather serve as Tools for Thought (TfT) that support and augment cognition? The goal is to enable better learning and higher-quality knowledge work while benefiting from the productivity gains promised by GenAI.

In this poster, I will present the process-oriented support framework from my doctoral research. The dominant paradigm when designing AI systems is automation-centric support, i.e., the AI automatically generates a complete solution, which the user then reviews. This style of interaction removes the user from the work process and forces them to reason backward from the result to the input, which is effortful and error-prone. Process-oriented support is an alternative to automation-centric support where the user remains fully engaged in their work process. The AI gives more incremental support that fits into this process, targeting intermediary stages of a task rather than fully solving it. This allows the user to reason forward from input to result. Essentially, the AI helps the user to solve the task themselves, rather than solving the task for the user.

Designing GenAI tools to emphasize forward reasoning like in process-oriented support could greatly facilitate users' ability to contribute their own reasoning to a given task, thereby enhancing learning and productivity. I will present empirical evidence from decision-making in aviation and finance, showing that process-oriented support tools can integrate more deeply with human reasoning and lead to better task outcomes than automation-centric support.



WHO IS HERE

08

Details of participants



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BIOGRAPHY

Niels Bartels studied alongside his professional career at the Baden-Wuerttemberg Cooperative State University (DHBW) Stuttgart and the University of Wuppertal. He received his Ph.D. from the Institute of Construction Management at TU Dresden with a dissertation entitled A Structural Model for Data Exchange in Facility Management.

Before joining academia, he worked as an Innovation Manager at GOLDBECK GmbH, where he was responsible for buildingSMART activities and led projects on the digitalization of construction processes and the systematization of building services engineering.

In August 2022, he was appointed Professor of Digital Planning and Construction at TH Köln, where he leads the teaching and research area of Digital Planning and Construction. His research focuses on Artificial Intelligence (AI) in the built environment, lifecycle-oriented data management, Building Information Modeling (BIM), digital transformation in the architecture, engineering and construction (AEC) industry, and reality capturing technologies for the digital representation and management of the built environment.

RESEARCH INTERESTS

- Artificial Intelligence in the built environment (Implementation, Use Cases)
- Smart Building and Digital Twins
- Data exchange across the life cycle,
- Reality Capturing of buildings (Photogrammetrie, 3D Gaussian Splatting (3DGS), Laserscanning)
- BIM and Sustainable Construction
- Metaverse for Architecture, Engineering, Construction and Operation Augmented Reality (AR), Virtual Reality (VR)

COLLABORATION IDEAS

- Artificial Intelligence for the Built Environment and cross-section exchange
- Lifecycle-Oriented Data Management and Digital Twins using sensors, collecting data, establishing data bases for AI training environments
- Reality Capturing and Building Information Modeling (BIM) based on AI applications



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BIOGRAPHY

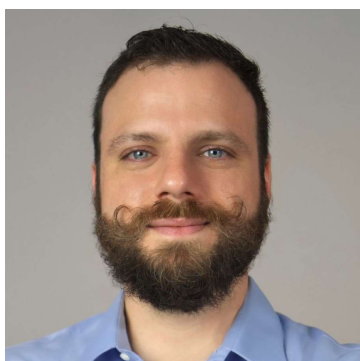
Born in 1991, Kevin earned two master's degrees in quantitative economics (specializing in Applied Microeconomics) and information systems (specializing in AI). He completed his PhD at Goethe University Frankfurt in 2018. Afterward, he worked as an AI specialist at Frankfurt's startup hub TechQuartier. In 2020, he joined the SAFE Leibniz Institute for Financial Research as a PostDoc, investigating the impact of AI on the financial system. From 2023 to 2024, he served as a Tenure Track Assistant Professor for E-Business and E-Government at the University of Mannheim. In 2025, Kevin returned to Goethe University Frankfurt as tenured Professor for Game-Theoretic and Causal AI, which is integrated into Hessian.AI. He also serves as a bridge professor to the Leibniz Institute for Financial research. His work has been published in leading journals such as Information Systems Research, Management Science, Marketing Science, and the Journal of the Association for Information Systems. He has received prestigious awards for his research, including the JAIS Best Paper Award, ISR Best Paper Award, the AIS Senior Scholar Award.

RESEARCH INTERESTS

Kevin Bauer's research lies at the intersection of technology and economics, with a focus on explainable AI and causal machine learning, and human information processing. His ongoing research projects feature field and lab experiments aiming to understand how AI systems and organizational structures need to be designed to enable effective human-AI collaboration. His work has been published in leading journals such as Information Systems Research, Management Science, Marketing Science, and the Journal of the Association for Information Systems.

COLLABORATION IDEAS

Field studies, optimization of human-AI collaboration, use cases for AI, agentic AI applications



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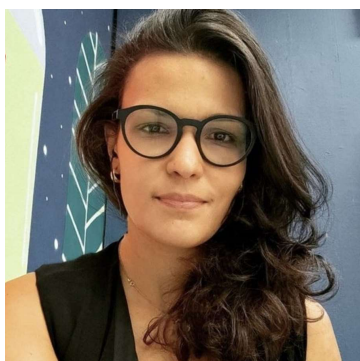
Dr. Benatto Perino is a postdoctoral researcher in the Bioprocess Engineering department at the University of Hohenheim, Germany. His research focuses on microbial biotechnology, recombinant protein production, and the development of sustainable bioprocesses using bacterial cell factories. He has extensive experience with *Bacillus subtilis* and *Escherichia coli*, particularly in the production of biosurfactants such as surfactin and recombinant milk proteins such as caseins. His work combines strain engineering, fermentation technology, process optimization, downstream processing, and analytical characterization. His contribution includes projects involving high-cell-density cultivation, oxygen-controlled bioprocesses, and the use of renewable substrates for biotechnological production. He is also involved in teaching, and student supervision. His scientific interests include synthetic biology, food biotechnology, sustainable biomanufacturing, and the development of microbial platforms for future food and bio-based applications.

RESEARCH INTERESTS

Microbial biotechnology, bioprocess engineering, and synthetic biology, with a focus on bacterial cell factories for sustainable production of biosurfactants and recombinant food proteins. Key interests include *Bacillus subtilis* strain engineering, fermentation optimization, oxygen-controlled bioprocesses, downstream processing, and future food biotechnology.

COLLABORATION IDEAS

Development of collaborative projects on microbial cell factories, sustainable bioprocesses, and recombinant food proteins. Potential topics include strain engineering, fermentation scale-up, biosurfactant production, downstream processing, valorization of renewable substrates, and synthetic biology approaches for future food and bio-based applications.



Natália Marroni **BORGES**

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■ BRAZIL

BIOGRAPHY

Natália Marroni Borges is a professor at the Federal University of Rio Grande do Sul (UFRGS), with a career that has run in parallel between academia and the corporate world. She holds a PhD in Information Systems from UFRGS and brings over 20 years of corporate experience in IT governance, innovation, and AI strategy and governance. She is a member of several research groups, including the IEA Future Lab, RAIES, and the ABES Think Tank, and co-founded NAVI, an AI innovation hub connecting academia and industry. Her research centers on AI governance in organizational contexts, with particular interest in foresight methodology, Shadow AI, and regulatory frameworks

RESEARCH INTERESTS

AI governance in organizational contexts, with applications across higher education, public administration, and the financial sector

Foresight methodology and strategic anticipation applied to digital transformation

The phenomenon of Shadow AI within organizations

Development of regulatory and institutional frameworks for AI

Social, ethical, and human-centered perspectives on AI and other emerging technologies, beyond a purely technocratic lens

COLLABORATION IDEAS

AI governance in organizational and institutional contexts

Generative AI institutionalization in higher education

AI governance in the financial sector

Shadow AI and organizational trust in AI systems

Foresight and regulatory anticipation



Jéssica Tainah da Silva **BOTELHO**

Volunteer Researcher at Imago – Research Group on Technological and Digital Images (Federal University of Amazonas)

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■ BRAZIL

BIOGRAPHY

PhD in Communication and Culture from the Federal University of Rio de Janeiro (UFRJ), journalist, and researcher focusing on communication and territory, particularly in the Amazon. Northern Region Coordinator for the News Atlas and Coordinator of the Center for Popular Communication and Audiovisual Production (CPA), a Manaus-based collective working at the intersection of climate justice, digital human rights, and diversity.

RESEARCH INTERESTS

Communication, territory, and socio-environmental issues in the Brazilian Amazon. Main interests: environmental and local journalism, socio-environmental disinformation, information integrity, digital rights, and responsible uses of generative AI. I am also interested in the socio-environmental impacts of data centers in Brazil and in how digital technologies can support knowledge circulation without reinforcing territorial inequalities or marginalising Amazonian perspectives

COLLABORATION IDEAS

I am interested in developing interdisciplinary Brazilian-German collaborations on responsible AI, information integrity, and the socio-environmental impacts of digital infrastructure. I hope to contribute an Amazonian perspective to discussions on AI governance, territorial inequalities, data centers, and the inclusion of local and traditional knowledge.



Anderson BRILHADOR

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■ BRAZIL

BIOGRAPHY

Anderson Brilhador is a professor and researcher at the Federal University of Technology – Paraná (UTFPR), Brazil. His work combines Artificial Intelligence, Machine Learning, Data Science, Computer Vision, and Software Engineering, with applications across real-world domains. He also develops intelligent software platforms and digital tools that connect research with practical applications.

RESEARCH INTERESTS

Artificial Intelligence; Machine Learning; Deep Learning; Computer Vision; Semantic Segmentation; Software Engineering; Intelligent Software Systems; Data Science; AI Applications; Digital Technologies; Agriculture and Rural Development.

COLLABORATION IDEAS

I am interested in collaborations between Brazilian and German institutions in Artificial Intelligence, Computer Vision, and intelligent software systems. I am particularly interested in applying these technologies to agriculture and other real-world challenges. Potential collaborations include joint research projects, AI and software development, shared datasets, student and researcher exchanges, joint publications, and international funding proposals.



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BIOGRAPHY

Anne Burkhardt is a postdoctoral researcher at the Center for Rhetorical Science Communication Research on Artificial Intelligence (RHET AI) at the University of Tübingen. Her current research focuses on AI and global (in)justice, with a particular emphasis on approaches and perspectives from the Souths. She explores how structural injustices—including infrastructural, environmental, epistemic, narrative, and engagement injustices—manifest and are reinforced in the context of AI development and use, and how they can be addressed.

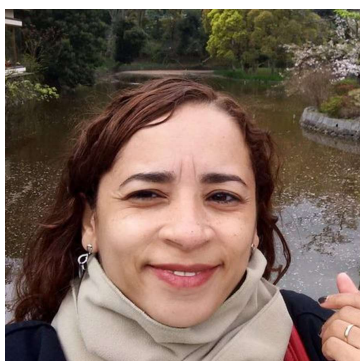
Anne Burkhardt studied Film and Theatre Studies at the University of Mainz. She received her PhD from the Institute of Media Studies at the University of Tübingen with a dissertation on cinematic representations of the Colombian internal conflict, for which she received the ADLAF dissertation award. Before joining the RHET AI Center, she worked as a research assistant at the International Center for Ethics in the Sciences and Humanities (IZEW).

RESEARCH INTERESTS

- Where and how do global injustices manifest in the context of AI, and what are pathways toward a globally just engagement with AI technologies?
- How do users from diverse cultural and social backgrounds perceive and navigate the persuasive power of LLM chatbots? How can they be prepared to build resilience against related uncertainties and risks?
- What are effective strategies for engaging diverse audiences with context-sensitive science communication and critical AI literacy, locally and globally?

COLLABORATION IDEAS

- To establish an interdisciplinary, intercultural research network on AI & Global Justice, fostering collaborative research, joint publications, and grant applications.
- To conduct interdisciplinary and intercultural research on the persuasive power of LLM chatbots.
- To co-create local and global frameworks for context-sensitive science communication about and with AI, as well as for strengthening critical AI literacy.



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BIOGRAPHY

Aline Paes is an Associate Professor at the Institute of Computing of the Universidade Federal Fluminense (UFF), Brazil, where she is a faculty member of the CAPES 7 Graduate Program in Computer Science and leads the Machine Learning and Language Learning (MeLLL) research group. She holds a Ph.D. in Systems and Computer Engineering from COPPE/UFRJ and completed a research internship at Imperial College London. She is a recipient of the CNPq Research Productivity Fellowship and the FAPERJ Young Scientist Award. She is also a researcher affiliated with three Brazilian National Institutes of Science and Technology (INCTs) and a member of Brasileiras em PLN (Brazilian Women in NLP). She serves as an Associate Editor for international journals and holds leadership roles within the Artificial Intelligence and Natural Language Processing communities.

RESEARCH INTERESTS

Machine Learning and Natural Language Processing, with an emphasis on multilingual and generative language models, trustworthy AI and AI for social impact, adaptation techniques for low-resource scenarios, figurative language processing, text simplification, reasoning with relational machine learning, and the detection of misinformation, hate speech, bias, propaganda, and online manipulation.

COLLABORATION IDEAS

Development and evaluation of large language models (LLMs), with a focus on multilinguality, reasoning, transparency, bias mitigation, and trustworthiness. I seek to develop joint research projects on the responsible use of generative AI in high-impact domains such as finance and healthcare, as well as on the creation of novel methods, benchmarks, and evaluation resources for trustworthy AI, fostering collaboration among academia, government, and industry.



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BIOGRAPHY

Dr. Binger Chen is a postdoctoral researcher at Humboldt University of Berlin. She received her PhD in Computer Science from Technical University Berlin and BIFOLD institute, with research focused on mining big code data and developing language models for automatic program transformation. Her current research focuses on AI agents, big data mining, trustworthy AI-human interaction, and earth observation workflow. She also has industry experience as an AI software developer at Porsche Digital and Samsung Electronics.

RESEARCH INTERESTS

My research focuses on trustworthy, data-centric agentic AI systems that combine LLMs with real-world data, structured knowledge, tools, and human feedback. I investigate how such systems can plan, execute, and optimize complex workflows (e.g., in software engineering and Earth observation), while remaining reliable, transparent, adaptable, and scientifically valid.

COLLABORATION IDEAS

Possible directions include generating realistic multimodal scientific workflow scenarios (such as earth observation workflow), creating synthetic workflow traces and failure cases for training and evaluation, and studying how synthetic and real data can improve tool selection, uncertainty estimation, and adaptive agentic workflows.



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BIOGRAPHY

Dr. Jan Cirullies is a professor at Dortmund University of Applied Sciences and Arts (FH Dortmund) since 2020, focusing on supply chain management and digital logistics. Previously, he led the Digitalization in Logistics department at the Fraunhofer Institute for Software and Systems Engineering ISST, where he remains a senior scientist. Before that, he headed a team for logistics simulation and decision support systems at Fraunhofer IML. Over his career, he has contributed to more than 40 publicly or industry-funded research projects. His current research explores (semi-)automated data model generation for simulation, leveraging machine learning and process mining to derive process models from shop floor events. These approaches are tested in industrial settings as well as in the Logistics Innovation Lab at FH Dortmund, which he co-founded in 2024. In the "LogILab", data from information systems and sensors is integrated to a digital twin as foundation for the application of various algorithms.

RESEARCH INTERESTS

Digital twins, logistics, production, resource planning, sensor data, real-time processing, process mining, artificial intelligence, discrete-event simulation, monitoring

COLLABORATION IDEAS

Fundamental research: Development of methods, particularly AI-based, for generating digital twins in production and logistics

Applied research: transfer projects with enterprises, particularly from industry



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BIOGRAPHY

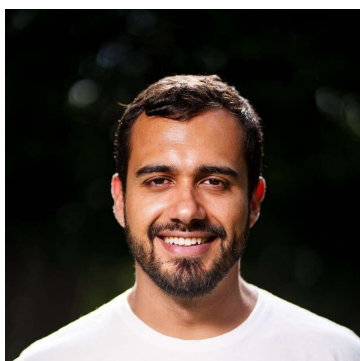
Associate Professor at the Federal University of Ceará (UFC), Quixadá Campus. She holds a Ph.D. in Computer Science from UFC (2022), including a doctoral research internship at the University of Versailles Saint-Quentin-en-Yvelines (France), and a Master's degree in Computer Science from UFC, with an academic exchange at the University of Alberta (Canada). She is a researcher at the Insight Data Science Lab (UFC), where she works on research, development, and innovation projects in partnership with public institutions, such as the Ministry of Justice, the Government of the State of Ceará, and the National Fund for Education Development (FNDE), as well as private companies. Her research focuses on Machine Learning, Artificial Intelligence, Spatiotemporal Data Processing, Natural Language Processing, and Digital Health. She currently leads the research track "Anomaly Detection in Biomedical Data and User Feedback" within the AI4Wellness project, sponsored by Samsung Brazil.

RESEARCH INTERESTS

Machine Learning applied to Digital Health. Natural Language Processing. Explainable AI

COLLABORATION IDEAS

Integrating predictive Machine Learning models and LLMs to produce more robust, transparent, explainable, and trustworthy decisions in data-driven applications.



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BIOGRAPHY

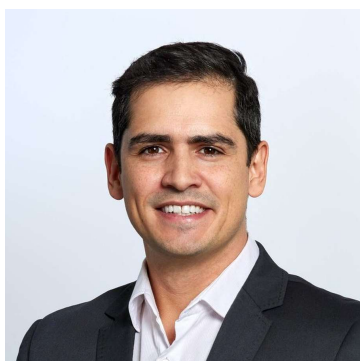
Washington Cunha is an Assistant Professor at the Institute of Computing, University of Campinas (UNICAMP), Brazil. He received his Ph.D. in Computer Science from the Federal University of Minas Gerais (UFMG), including a research internship at the Institute of Information Science and Technologies of the National Research Council of Italy (ISTI-CNR). His research focuses on Natural Language Processing, Information Retrieval, Machine Learning, and Responsible and Sustainable Artificial Intelligence, with particular interests in efficient learning methods, large language models, and text classification. He has published in leading international venues such as SIGIR, ICTIR, and ECIR, and has collaborated with academic and industrial partners, including Google Research and NVIDIA.

RESEARCH INTERESTS

Natural Language Processing, Information Retrieval, Machine Learning, Large Language Models (LLMs), Responsible AI, Sustainable AI, Data-Centric AI, Instance Selection, Retrieval-Augmented Generation (RAG), Trustworthy AI, Multilingual NLP, Text Classification.

COLLABORATION IDEAS

I am interested in establishing collaborations that combine expertise in NLP and AI to address challenges in efficient, trustworthy, and sustainable AI. In particular, I seek opportunities to collaborate on data-centric approaches for LLMs, multilingual NLP, RAG, and AI evaluation. I also welcome interdisciplinary collaborations that promote open science, and foster international research projects, student exchanges, and joint funding initiatives between Brazil and Germany.



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BIOGRAPHY

Denilton Luiz Darold has been a researcher at Fraunhofer ISI since 2020 in the Department of Innovation and Knowledge Economy. He holds a doctorate in Economics with a focus on Innovation Management from the University of Hohenheim and lectures in the Data Science Master's Program at SRH Heidelberg. He holds a bachelor's degree in Computer Science and an M.Sc. from the State University of Santa Catarina (UDESC), Brazil. With extensive experience as a software developer specializing in Big Data applications, his research focuses on measuring and assessing innovation through metrics derived from novel data sources and data science methods, with a particular emphasis on innovation economics.

RESEARCH INTERESTS

My research focuses on applying data science and AI to large-scale structured and unstructured data to study innovation, technology adoption, and economic impact. I develop data-intensive methods for research and policy analysis, including data engineering, NLP, machine learning, and econometric approaches. Current interests include generative AI, research infrastructures, AI evaluation, and the use of data-driven methods to address societal and innovation challenges.

COLLABORATION IDEAS

I am interested in collaborations combining AI, data science, and domain expertise to address challenges in innovation, sustainability, biodiversity, and economic development. Potential topics include large-scale data integration, AI adoption and impact assessment, trustworthy and explainable AI, research infrastructures, and policy-oriented analytics. I particularly welcome German–Brazilian collaborations that connect methodological advances with real-world applications.



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BIOGRAPHY

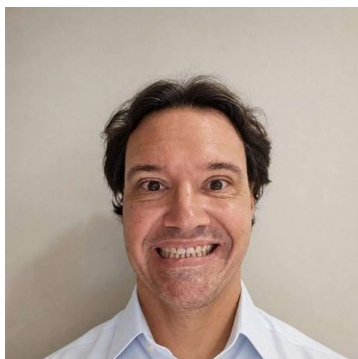
Dr. Michael Dunst is a consultant, researcher, and lecturer specializing in innovation management, quality management, organizational development, and artificial intelligence in business contexts. He holds a Ph.D. from TU Berlin and Esslingen University of Applied Sciences, where his research focused on synergies between innovation and quality functions in product development. He earned degrees in Innovation Management and Industrial Engineering, complemented by international study experiences in Japan and the United States. Since 2017, he has worked as a Senior Consultant and lecturer in the Master's program in Innovation Management at Esslingen University of Applied Sciences. He is also part of the scientific leadership of AnalyzES!, the regional AI Lab at Esslingen University, which supports companies in identifying and applying AI potential. Dr. Dunst has authored publications on innovation and quality management and has led applied research projects.

RESEARCH INTERESTS

Dr. Michael Dunst's research focuses on innovation management, quality management, and the application of Artificial Intelligence in organizations. His work explores how innovation, quality, and AI capabilities can be integrated to improve product development, decision-making, and organizational performance. Additional interests include strategic innovation management, change management, technology commercialization, digital transformation, and AI-driven analytics for business and industrial applications.

COLLABORATION IDEAS

Collaboration opportunities include joint research projects on innovation management, quality management, and the application of artificial intelligence in industrial environments. Particular interests are AI-driven innovation processes, digital transformation, organizational development, and industry–university cooperation. The goal is to develop practical solutions that strengthen competitiveness, sustainability, and innovation capabilities across international contexts.



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■ BRAZIL

BIOGRAPHY

I have a degree in electrical engineering from the State University of Londrina, with a master's and a PhD in dynamic systems from the School of Engineering of São Carlos.

As a Post-doctorate reserach in AI at UFG, I worked on projects applying LLMs in the legal field and data mining, as well as optimization applications for logistics.

I have experience in the electric power distribution field, having worked in operational service areas, operation, and maintenance, before moving on to process development and engineering support.

I currently work as the executive director of the Artificial Intelligence Center of Excellence in Artificial Intelligence (CEIA/UFG), and I'm also responsible for research related to Digital Twins, Robotics, and applications of LLMs in real-world problems.

RESEARCH INTERESTS

Digital Twins, Robotics, Real-world LLM applications, Agentic AI, Power distribution systems

COLLABORATION IDEAS

My main interests are related to discussing possible collaborations in the field of digital twin developments in Brazilian production processes based on the German experience. Additionally, I'm looking for collaborations in research related to the application of agent-based AI in critical infrastructure sectors in Germany, mainly focusing on power electrical systems.



Vagrant GAUTAM

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■ GERMANY

BIOGRAPHY

Vagrant Gautam is an Independent Postdoctoral Fellow at the Heidelberg Institute for Theoretical Studies, where they study both social and technical aspects of trustworthy natural language processing – especially fairness, reasoning and robustness. On the technical side, they propose methods for and evaluations of these aspects that draw from the social sciences and humanities. On the social side, they investigate gaps and opportunities in how we define and operationalize abstract concepts like trustworthiness, gender, etc., as well their societal implications. They hold a PhD in computer science from Saarland University, before which they worked on speech recognition, discourse processing, and experimental phonology in academia and industry.

RESEARCH INTERESTS

Trustworthy natural language processing, fairness, reasoning, interpretability, linguistics

COLLABORATION IDEAS

A framework for trustworthy AI



Arlindo GALVÃO

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■ BRAZIL

[LinkedIn](#) | [ResearchGate](#)

BIOGRAPHY

CEO of the Competence Center in Immersive Technologies. Director of Internationalization of the Center of Excellence in Artificial Intelligence. He has a career of innovation with a solid academic background (cum laude), a Bachelor's degree in Computer Engineering from PUC Goiás, a Master's degree in Electronic and Computer Engineering from ITA, and a PhD in Computer Science from UFG. He leads innovation projects in healthcare, energy, logistics, and other sectors. Most of these projects are sponsored by public and private companies such as Petrobras, Natura, the National Electric Energy Agency, the Ministry of Health, Fiocruz, and Americas Health Group, among others. He is a board member of technology-based startups. He is a Full Professor of Natural Language and Speech Processing at the Federal University of Goiás, where he is a permanent member of the Master's and Doctoral programs in Computer Science.

RESEARCH INTERESTS

As a professor at the Institute of Informatics at the Federal University of Goiás (UFG), my research is centered on Artificial Intelligence, with a primary focus on Machine Learning, Deep Learning, and Pattern Recognition. I am passionate about applying these technologies to solve complex challenges, particularly within Natural Language Processing (NLP) and Audio & Voice Processing.

My work extends to impactful areas like digital health, where I develop systems for medical diagnosis and assistive technologies, and I am also deeply engaged with Generative AI and immersive technologies. A core part of my mission is to bridge the gap between academia and industry. Through my role at the Center of Excellence in AI (CEIA), an Embrapii unit, I lead collaborative R&D projects with companies, drive innovation, and foster the development of highly skilled professionals, always seeking to build strong International partnerships.

COLLABORATION IDEAS

I am interested in establishing international collaborations that connect academia, industry, and innovation ecosystems, particularly in Artificial Intelligence, Natural Language and Speech Processing, Generative AI, immersive technologies, and digital health. Potential collaborations may include joint R&D projects, researcher and student exchanges, co-development of applied AI solutions, and partnerships with companies and research institutions, with a strong focus on translating scientific advances into real-world impact across healthcare, energy, logistics, and other strategic sectors.



Priscila GONSALES

Research Associate and Consultant

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■ BRAZIL

BIOGRAPHY

Researcher in critical AI literacies, I work at the intersection of technology, education and social equity. PhD in Languages and Technologies from UNICAMP. MA from the Programme in Intelligence Technologies and Digital Design at PUC-SP. Honorary Research Associate at the Centre for Sociodigital Futures at the University of Bristol (UK). Consultant for UNESCO, and an associate of the UNESCO Chair in Open Education and Technologies for the Common Good (UnB) and of the Centre for Research in Post-humanism and Digital Humanities at IEL/UNICAMP. My research focuses on building sociodigital futures that are fairer, more democratic and more widely shared. As director of Educadigital, I was responsible for introducing the Design Thinking (DT) approach for educators in Brazil in 2014, connecting pedagogical innovation with social transformation. More recently, I have integrated DT into training and facilitation processes involving serious games, such as the Talents Cards deck.

RESEARCH INTERESTS

Critical AI Literacies
Sociodigital Futures
Platformisation and Datafication of Education
Open Education and Digital Rights
Digital Citizenship and Media Literacy
Teacher's Professional Development

COLLABORATION IDEAS

I can contribute to the topic of AI literacies from a critical post-humanist perspective, drawing on my research into distributed cognition and cognitive assemblages. I can also present the serious game "Cartas dos Talentos" (Talent Cards), which has an English version and is one of the research and training methods for sociodigital futures.



Matteo GROSSE-KAMPMANN

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■ GERMANY

BIOGRAPHY

Prof. Dr. Matteo Große-Kampmann received his PhD from Ruhr University Bochum in 2022 in cooperation with the Westfalian University of Applied Sciences. He was appointed Professor of Distributed System at Rhine-Waal University of Applied Sciences in 2023. He is vice dean of the faculty of communication and environment. His research focuses on security, private and safe systems as well as software education in the age of artificial intelligence.

RESEARCH INTERESTS

My research interests are located in the security, privacy and safety of systems, ranging from technical analyses of applications and systems to linguistic analyses in the case of disinformation. In the case of disinformation I journey towards the emerging cognitive AI boundary and how disinformation will influence future generations.

COLLABORATION IDEAS

I would like to collaborate on the convergence of BCI, Cognition and AI model as well as the impact of disinformation in these systems. There is currently no threat model available for a system that combines these three technologies. Furthermore, little is known about the perception of the public on these new threats. My interest would be to collaborate on these issues.



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■ GERMANY

BIOGRAPHY

Marc Hasselwander is a Postdoctoral Researcher and a Lecturer with the Technical University of Berlin. He holds a Ph.D. degree from the University of Coimbra through the MIT Portugal Program. He has previous stints as a Visiting Researcher with the University of the Philippines, in 2019, and the University of Oxford, in 2025.

Dr Hasselwander has presented his research at international conferences and published in top tier journals, such as Journal of Retailing and Consumer Services, Sustainable Cities and Society, Environmental Innovation and Societal Transitions, European Research on Management and Business Economics, and Transportation Research Part A: Policy and Practice.

RESEARCH INTERESTS

His research interests include the intersection of consumer behavior and technology adoption, informed by perspectives from business/marketing, information systems, and psychology. In his most recent publications, he has focused on agentic AI, AI agents, digital platforms, and super apps.

COLLABORATION IDEAS

I am interested in cross-country studies and experiments – particularly comparing Germany and Brazil, and potentially including additional countries – that investigate how consumers across different cultures adopt and use agentic AI systems. Relevant application areas include higher education, e-commerce, urban mobility, and other consumer-facing domains.



Felix HOLL

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■ GERMANY

BIOGRAPHY

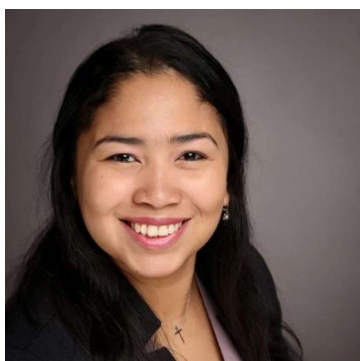
Dr. Felix Holl is a postdoctoral researcher at the Africa Institute, Neu-Ulm University of Applied Sciences, Germany. His research focuses on digital public health, global health informatics, artificial intelligence, and resilient health systems, with a particular emphasis on implementing and evaluating digital innovations in low- and middle-income countries. He has led and contributed to numerous international collaborations across Africa, Europe, and North America, working with universities, ministries of health, humanitarian organizations, and international partners. Beyond his research, he serves as Editor-in-Chief of *BMJ Connections Digital Health & AI*, Associate Editor of *npj Digital Public Health*, and chairs the Global Health Informatics Working Group of the American Medical Informatics Association (AMIA). His work aims to generate evidence that enables the equitable, sustainable, and scalable adoption of digital technologies to strengthen health systems worldwide.

RESEARCH INTERESTS

Digital public health; global health informatics; artificial intelligence in health; implementation science; digital health policy and governance; health information systems; evaluation of digital interventions; health system resilience; humanitarian health informatics; digital transformation of health systems; equitable AI implementation in low- and middle-income countries.

COLLABORATION IDEAS

Interested in interdisciplinary collaborations on AI and digital technologies for resilient health systems, implementation science, digital health governance, and public health preparedness. Particularly keen to develop joint projects combining computer science, engineering, health sciences, and social sciences to translate innovative technologies into scalable, equitable, and sustainable solutions with real-world impact.



Isabel HÜBENER

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■ GERMANY

BIOGRAPHY

Prof. Dr. Isabel Hübener has been Professor of Applied Artificial Intelligence at the Deggendorf Institute of Technology (DIT), Germany, since 2025. She studied Physics and Astronomy in Colombia before earning her PhD in Astrophysics from the University of Potsdam in 2013. At the Leibniz Institute for Astrophysics Potsdam (AIP), she worked on the analysis of large-scale cosmological data from the WMAP satellite, developing expertise in data analysis, statistical modeling, and uncertainty quantification. She later transitioned into artificial intelligence, machine learning, and deep learning, coordinating interdisciplinary research projects at the University of Kassel. Afterwards, she moved from academia to industry, where she worked for six years as a Data Scientist, developing AI and machine learning solutions for real-world industrial applications. Her combined academic and industrial experience now shapes her teaching and research in applied AI.

RESEARCH INTERESTS

Machine Learning, Context recognition, Feature Selection, Computer Vision, Anomaly detection

COLLABORATION IDEAS

I am particularly interested in collaborations that combine AI with scientific discovery. Coming from a background in astrophysics and industry, I enjoy developing machine learning methods that help understand complex systems and solve real-world problems. I would be excited to collaborate with Brazilian researchers on applications in robotics, environmental sciences, agriculture, healthcare, or astronomy, while also advancing the underlying AI methods.



Deise Aline KNOB

Justus Liebig University Giessen - Organic Farming

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■ GERMANY

BIOGRAPHY

Dr. Deise Aline Knob is a Postdoctoral Researcher at the Department of Organic Farming, Justus Liebig University Giessen (JLU), Germany. She holds a Ph.D. in Animal Science and conducts research on sustainable and climate-resilient livestock production, with a particular focus on dairy systems. Her work integrates animal nutrition, precision livestock farming, environmental assessment, and emerging digital technologies to improve animal health, productivity, resource-use efficiency, and environmental sustainability. She collaborates in international research projects on organic farming and integrated crop-livestock systems, contributing to the development of science-based solutions for more sustainable and resilient agricultural systems.

RESEARCH INTERESTS

Sustainable and climate-resilient livestock production; precision livestock farming; digital agriculture; AI applications in livestock systems; integrated crop-livestock systems; sensor-based animal monitoring; Internet of Things (IoT) and LoRaWAN networks; environmental assessment; greenhouse gas emissions; virtual fencing; computer vision; data-driven decision support for sustainable agriculture.

COLLABORATION IDEAS

Open to interdisciplinary collaborations with researchers developing AI and digital technologies for real-world applications. Our experimental dairy farm provides an ideal environment to test and validate emerging solutions, including sensor technologies, computer vision, IoT, and digital monitoring systems, using high-quality reference measurements under practical farming conditions..



Anna KRUSPE

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■ GERMANY

BIOGRAPHY

Prof. Dr. Anna Kruspe is a research professor at the Faculty of Computer Science and Mathematics, Hochschule München. She completed her PhD at Fraunhofer IDMT/TU Ilmenau in 2018, researching automatic speech recognition in singing, following research stays at Johns Hopkins University, USA, and AIST, Japan. She then worked at the German Aerospace Center (DLR) in Jena, leading a Machine Learning group, before joining TU München as deputy head of the "Data Science in Earth Observation" chair. She held a professorship in Natural Language Understanding at TH Nürnberg before joining Hochschule München in 2024, where she now researches bias, fairness, and geocultural representation in AI, including geographic and geocultural biases in LLMs and text-to-music generation.

RESEARCH INTERESTS

My research centers on fairness, bias, and trustworthiness in generative AI, with a focus on how language models and text-to-music systems reproduce geocultural, linguistic, and regional stereotypes. I am extending this empirical work toward mechanistic interpretability, investigating where in a model geocultural bias is encoded, and toward uncertainty quantification as a complementary signal for bias detection. A second strand explores the intersection of Music Information Retrieval and NLP, including text-based interfaces for music, lyrics retrieval, and language in music description. I am also interested in synthetic data as a fairness tool for closing representation gaps, and in copyright and research ethics questions around AI-generated art.

COLLABORATION IDEAS

I see strong links to Aline Macohin's work on benchmark gaps for culturally and linguistically situated harms in Portuguese, my geocultural bias benchmark could be extended to Brazilian contexts. Francielle Vargas's attention-alignment interpretability connects to my mechanistic work on locating bias in model internals. Beyond the session, I would welcome a joint DAAD or Brazilian-German funding application on geocultural bias in generative AI, potentially involving Aline Paes, Thiago Naves, and Isabel Huebener.



Christian LEITERER

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■ GERMANY

BIOGRAPHY

Christian Leiterer is Innovation Manager at Ernst-Abbe-Hochschule Jena and Friedrich Schiller University Jena, where he develops strategic initiatives at the interface of AI, digitalisation and research. Trained as a biochemist, he holds a doctorate from the Leibniz Institute of Photonic Technologies and held research positions at Macquarie University Sydney (DAAD P.R.I.M.E./Marie Curie) and UC San Diego before moving from the lab into innovation management. He is an Honorary Adjunct Fellow at Macquarie's School of Engineering.

His work bridges technology and application: he co-ed the acquisition of three endowed professorships, including one on Explainable AI, builds AI prototypes such as OCR-LLM pipelines and locally hosted language models, and has delivered more than 30 workshops on AI for universities, foundations and SMEs in Europe and Australia. At BRAGFOST he presents a hybrid, open and sustainable approach to AI-based post-OCR correction with multimodal foundation models.

RESEARCH INTERESTS

Christian Leiterer works to make generative AI usable, transparent and sustainable. He develops hybrid pipelines combining OCR with multimodal foundation models for post-OCR correction, and locally hosted LLMs that keep sensitive data under institutional control. A second focus is transfer: how universities, foundations and SMEs adopt AI critically, not blindly. Coming from biochemistry and photonics, he is drawn to problems where a technical solution only works if the community around it does.

COLLABORATION IDEAS

Christian Leiterer is looking for partners holding text corpora that current OCR pipelines handle poorly. Portuguese-language, historical or administrative collections could be tested whether his approach transfers beyond German material. He is equally interested in joint work on locally hosted LLMs for institutions with limited compute or strict data requirements, and in exchanging workshop formats that bring AI into SMEs. A CONNECT working visit in either direction would be a natural first step.



Leopoldo André Dutra LUSQUINO FILHO

(Assistant Professor in the Department of Control and Automation Engineering at the Institute of Science and Technology of Sorocaba, UNESP)

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■ BRAZIL

BIOGRAPHY

Leopoldo Lusquino Filho is a professor at Unesp and a researcher at the Recod.ai lab (Unicamp). He holds a bachelor's degree in Computer Science, as well as a master's degree and a PhD in Systems and Computer Engineering, all from UFRJ. He completed his postdoctoral research at Unicamp's Institute of Computing. He currently coordinates LINCE, the AI academic league at Unesp. His research focuses on Artificial Intelligence applications for combating disinformation, predicting extreme weather events and epidemiological scenarios, cognitive robotics, and procedural content generation in virtual games.

RESEARCH INTERESTS

Complex network analysis, synthetic reality detection, time-series analysis, procedural content generation, meta-learning.

COLLABORATION IDEAS

Detection of synthetic disinformation content, AI-driven analysis of the evolution of disinformation narratives on social media, analysis of disinformation provenance, and analysis of cognitive dissonance formation and behavioral anomalies in AI agents exposed to disinformation.



Aline MACOHIN

PUCPR Affiliation (Graduate Professor) and Responsible AI (Website)
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■ BRAZIL

BIOGRAPHY

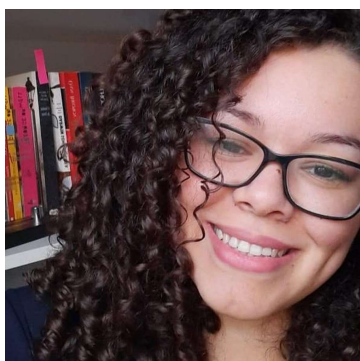
Aline Macohin is a Brazilian researcher with an interdisciplinary background in Law and Computer Science. She combines academic research with extensive professional experience as a systems analyst in the Brazilian public sector, where she works on Artificial Intelligence governance, cybersecurity, digital trust, and anti-fraud governance. She is also a postgraduate lecturer at the Pontifical Catholic University of Paraná (PUCPR), teaching courses on Artificial Intelligence and Law. Her doctoral research focuses on algorithmic transparency, explainability, accountability, and trustworthy AI in public-sector decision-making, bridging legal requirements with technical approaches to responsible AI. Her work seeks to operationalize AI governance through practical mechanisms that enhance transparency, auditability, and meaningful human oversight in high-impact AI systems.

RESEARCH INTERESTS

Artificial Intelligence Governance; Trustworthy AI; Explainable AI (XAI); Algorithmic Transparency; AI Accountability; Responsible AI; Public Sector AI; AI Regulation; AI Risk Governance; AI Auditing; Digital Government; Public Administration; AI Ethics; Machine Learning Interpretability; Large Language Models; Fairness; AI Safety.

COLLABORATION IDEAS

I am interested in collaborating with researchers working on trustworthy AI, explainable AI, AI governance, and public-sector AI. My research brings together computer science, law, and public administration to develop practical approaches to transparency, accountability, and meaningful human oversight in high-impact AI systems. I also welcome collaborations on large language models (LLMs), AI assurance, and the responsible deployment of AI in public institutions.



Helena de Almeida MAIA

Assistant Professor, Institute of Computing/UNICAMP

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■ BRAZIL

BIOGRAPHY

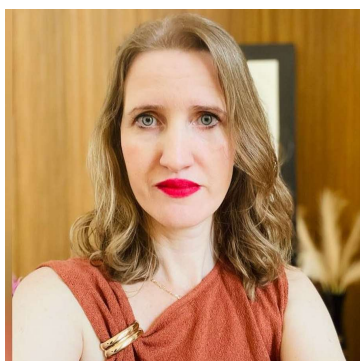
Helena Maia is an Assistant Professor at the Institute of Computing (IC) of the University of Campinas (UNICAMP). She is a researcher at the Artificial Intelligence (AI) Lab, Recod.ai, and Hub for Artificial Intelligence and Cognitive Architecture (H.IAAC). She holds Bachelor's degrees in Exact Sciences and Computer Science, a Master's degree from the Universidade Federal de Juiz de Fora (UFJF), and a Ph.D. in Computer Science from Universidade Estadual de Campinas (UNICAMP).

RESEARCH INTERESTS

My research interests include Ethics and AI Literacy, Natural Language Processing and Computer Vision. I am a researcher in the LIA (Human Language and Artificial Intelligence) project, conducted in collaboration with Applied Linguistics researchers. It focuses on the meaning-making process in multimodal models. Additionally, my work involves raising awareness about the impacts of artificial intelligence through games and lectures, targeting both university audiences and the 50+ demographic.

COLLABORATION IDEAS

Culturally-aware multimodal models; Responsible development and use of surveillance systems; Ethics and AI Literacy; Generative AI in research;



Alessandra MAIESKI

Federal University of Mato Grosso / Institute of Education / (Professor, Department of Theory and Foundations of Education)

■ BRAZIL

BIOGRAPHY

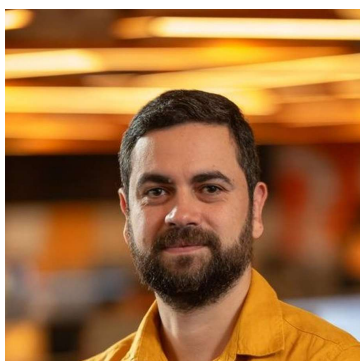
Professor at the Federal University of Mato Grosso (UFMT), affiliated with the Institute of Education (IE) and the Department of Theory and Foundations of Education (DTFE). Accredited faculty member of the Graduate Program in Education (PPGE/UFMT). She holds a Ph.D. and a Master's degree in Education from PPGE/UFMT, a Graduate Certificate (Specialization) in Information Technology in Education from UFMT, and a Bachelor's degree in Pedagogy. She is currently pursuing a second undergraduate degree in Artificial Intelligence Technology. She is a researcher at the Laboratory for the Study of Information and Communication Technologies in Education (LêTece), an affiliated member of the Center for Open and Distance Education (NEAD/IE/UFMT), a member of the Research Network on Democracy and Artificial Intelligence in Education (ReDIAE), and a researcher with the Innovation Network for Hybrid Education (RIEH/UFAL).

RESEARCH INTERESTS

My research interests focus on Learning Ecologies, Artificial Intelligence in Education, Digital Culture, Digital Education, Educational Technologies, Teacher Education, Higher Education, and Distance Education. My current research explores how Artificial Intelligence is reshaping learning ecologies in educational contexts. I am also involved in collaborative research on democracy and Artificial Intelligence in Education, hybrid education, and the digital transformation of higher education.

COLLABORATION IDEAS

International collaboration on Artificial Intelligence in Education; Learning Ecologies; teacher education; digital transformation of higher education; including comparative research; collaborative projects; joint publications.



Rafael Ferreira Leite de MELLO

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■ BRAZIL

BIOGRAPHY

Rafael Ferreira Mello is an Associate Professor at UFRPE and Senior Researcher at CESAR (Ph.D. from UFPE, postdoc from Edinburgh) who works on AI in Education, focusing on NLP and Large Language Models. He has led Brazilian Ministry of Education projects to improve student writing and build Learning Analytics dashboards for education policy, collaborated with partners like Google and institutions across Europe, Australia, and Latin America, and published 250+ papers. He also serves as an assistant editor and reviewer for venues including the Journal of Learning Analytics and the LAK conference.

RESEARCH INTERESTS

Learning Analytics, Small Language Models, Human Centered AI, Natural Language Processing

COLLABORATION IDEAS

I work at the intersection of NLP and Learning Analytics, focusing on Small Language Models as a cheaper, privacy-preserving alternative to large models for under-resourced settings like the Global South. I'm seeking partners to build and evaluate SLMs for writing feedback, discourse analysis, and policy support, plus Human-Centered AI tools co-designed with teachers and students. Open to joint research, co-supervision, exchanges, and grant proposals.



Carlos Vinicius Gomes **MELO**

EduWellTech — Founder, Director and CEO; University of São Paulo (USP) — Collaborating Professor, Graduate Program in Social Psychology, Department of Social and Work Psychology, Institute of Psychology

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■ BRAZIL

BIOGRAPHY

I am a psychologist with a PhD in Social Psychology from the University of São Paulo (USP). I am the founder, director and CEO of EduWellTech and a collaborating professor in USP's Graduate Program in Social Psychology. My work focuses on social psychology, racial and ethnic relations, student mental health, psychosocial governance and responsible AI. I develop approaches that integrate psychometrics, data science, artificial intelligence, participatory methods and socioenvironmental mapping to support well-being, belonging, student persistence and evidence-informed decision-making in higher education. My research has been supported by FAPESP and CNPq. I currently conduct research on AI-mediated persuasion, critical AI literacy, epistemic justice, democratic communication and mental health. I co-chair the BRAGFOST 2026 session "Generative AI for Next-Generation Education, Research, and Knowledge Transfer."

RESEARCH INTERESTS

My research focuses on social psychology, social cognition, racial and ethnic relations, student mental health, and responsible AI. Planned activities involve study how generative AI affects judgment, knowledge production, and well-being, by critical AI literacy and psychosocial impact frameworks.

COLLABORATION IDEAS

I seek Brazil–Germany collaborations on generative AI, education, psychosocial effects, well-being, and epistemic justice through comparative studies, shared methods, joint publications, researcher exchanges, workshops, and funding proposals.



Mariana Recamonde **MENDOZA**

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■ BRAZIL

BIOGRAPHY

Mariana Recamonde Mendoza is an Associate Professor at the Institute of Informatics of the Federal University of Rio Grande do Sul (UFRGS), Brazil, and a Research Professor at Hospital de Clínicas de Porto Alegre (HCPA). She is a CNPq Research Productivity Fellow and a faculty member of the Graduate Program in Computer Science (PPGC/UFRGS). She is passionate about developing and applying AI to address healthcare challenges and improve health outcomes. She leads the MLAB – Machine Learning & Applications in Biomedical Data research laboratory and coordinates research projects funded by Brazilian and international agencies. She has co-authored more than 100 peer-reviewed scientific publications and has supervised more than 90 undergraduate and graduate students in machine learning, bioinformatics, data science, and computing for healthcare. In 2026, she serves as Program Committee Chair of BRACIS, Brazil's flagship scientific conference on Artificial Intelligence.

RESEARCH INTERESTS

Her research focuses on trustworthy AI for healthcare. Her current research interests include predictive modeling for healthcare, multimodal biomedical data integration, algorithmic fairness and bias mitigation, synthetic data generation, explainable AI, methodological rigor in the development of reliable AI models, and computational approaches to epidemiological surveillance.

COLLABORATION IDEAS

I am interested in interdisciplinary collaborations on trustworthy AI for healthcare. My research interests include algorithmic fairness, synthetic data generation for bias mitigation, explainable AI, reliable machine learning, multimodal biomedical data integration, and computational approaches for public health and epidemiological surveillance. I am especially keen to build interdisciplinary collaborations that connect methodological advances in AI into real-world healthcare applications.



Caroline MENEZES

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■ GERMANY

BIOGRAPHY

Caroline Menezes holds a PhD in History and Theory of Art from Chelsea College of Arts, University of the Arts London, and is Co-Manager of the Institute for Digital Transformation and Innovation at BSP Business and Law School in Berlin. Her interdisciplinary research focuses on digital transformation and creative technologies, bridging academic enquiry with professional practice. Her work has been supported by international fellowships, including the German Chancellor Fellowship of the Alexander von Humboldt Foundation, through which she developed a research project at ZKM | Center for Art and Media Karlsruhe. With over a decade of experience as an art curator, writer and cultural manager, she has contributed to international publications and projects at the intersection of art, technology and digital culture.

RESEARCH INTERESTS

My research focuses on creative technologies, artificial intelligence and digital transformation, with particular attention to human-machine collaboration, creative agency and authorship. I also examine modern and contemporary art, exhibition histories, curatorial practices and transnational, non-Eurocentric approaches to art and digital culture.

COLLABORATION IDEAS

I would welcome interdisciplinary collaborations between AI researchers, educators, artists and social scientists to explore how creative technologies and generative AI can support knowledge transfer, foster human-centred innovation and incorporate culturally diverse, non-Eurocentric perspectives into digital transformation.



Filip MILETIC

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■ GERMANY

BIOGRAPHY

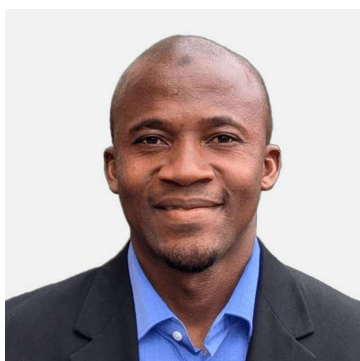
Filip Miletic is a postdoctoral researcher at the Institute for Natural Language Processing (IMS), University of Stuttgart. He holds a PhD in Linguistics from the University of Toulouse (France), with an interdisciplinary background in sociolinguistics and natural language processing (NLP). In general terms, Filip's research analyzes social and linguistic behaviors in connection with NLP technologies, and is motivated both descriptively (obtaining empirical insights) and methodologically (developing and evaluating data and models). This human-centric perspective on language technologies has led to publications in top-tier NLP as well as sociolinguistic venues; it is also at the core of Filip's teaching activities, both at the IMS and in broader initiatives (most recently at the ESSLLI summer school in 2024). Filip is actively involved in the international research community, in particular as a reviewer and area chair for major NLP conferences.

RESEARCH INTERESTS

Filip's main areas of interest are socially grounded computational modeling of linguistic phenomena (e.g., sociodemographic and temporal language variation); socially and linguistically motivated analysis of language technologies (e.g., detection and mitigation of bias in LLMs); and resource creation and evaluation (e.g., corpora with rich sociodemographic information). His recent work has increasingly focused on high-stakes domains such as scientific communication.

COLLABORATION IDEAS

Possible collaborations include interdisciplinary work with experts from other disciplines who share an interest in responsible AI systems, for example in terms of their robustness in interactions with members of different sociodemographic groups. Specific directions in Filip's ongoing work include assessing the effects of AI writing assistants on scientific communication, as well as sociolinguistically grounded analyses of LLMs as a source of linguistic knowledge in educational scenarios.



Hamza MOLUH NJOYA

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■ GERMANY

BIOGRAPHY

Hamza Moluh Njoya is a researcher at the Leibniz Centre for Agricultural Landscape Research (ZALF), Germany. He holds a PhD in Economics and Governance of Agriculture, Food, and Natural Resource Management from Humboldt University of Berlin and a Master of Science in Integrated Rural Development from the University of Dschang, Cameroon. His research focuses on how socioeconomic constraints, institutional arrangements, and policy incentives shape the adoption of environmental practices and the outcomes of restoration interventions. Using quantitative methods, including quasi-experimental approaches such as endogenous switching regression and propensity score matching, he analyses the causal effects of land restoration programs on household welfare, resource-use decisions, and technology adoption. His work contributes to environmental and development economics by providing evidence to inform the design of effective and inclusive policies for sustainable land management. He has conducted extensive field research in Togo, Côte d'Ivoire, and Cameroon.

RESEARCH INTERESTS

Forest landscape restoration; knowledge integration; artificial intelligence for sustainability; environmental governance; sustainable land management; food security; smallholder farming systems; socio-ecological systems; stakeholder engagement; interdisciplinary research; evidence synthesis; collaborative decision-making; sustainability transitions; policy analysis; ecosystem services.

COLLABORATION IDEAS

I am interested in exploring how generative AI can support knowledge integration for Forest Landscape Restoration (FLR) by connecting scientific, local, institutional, and policy perspectives. Potential collaboration topics include stakeholder knowledge needs, AI-assisted evidence synthesis, responsible AI, and context-sensitive decision-support systems that enable more collaborative and inclusive restoration planning in smallholder landscapes.



Dominik NAEHER

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■ GERMANY

BIOGRAPHY

Dominik Naeher is a Senior Research Fellow (Postdoc) at the University of Göttingen, where he also served as Interim Professor (W2) of Development Economics from 2024 to 2025. Before this, he worked as Assistant Professor of Economics at University College Dublin and earned his PhD from Goethe University Frankfurt (GSEFM) in 2018. His research combines economic modeling and empirical methods to study questions in development economics, information economics, and sustainability and environmental economics. A central focus is the determinants and consequences of technology adoption, including the role of information frictions in shaping adoption decisions and economic outcomes, and how technologies that mitigate these frictions transform individual behavior, markets, and public policy. His work has appeared in leading economics journals including *International Economic Review*, *RAND Journal of Economics*, and *Journal of Development Economics*. He has taught at universities in Germany, France, Ireland, China, Korea, and South Africa, and regularly provides policy advice to international organizations, including the World Bank, EBRD, ADB, GIZ, and UN agencies.

RESEARCH INTERESTS

Determinants and consequences of technology adoption, including the role of information frictions in shaping adoption decisions and economic outcomes, and how technologies that mitigate these frictions transform individual behavior, markets, and public policy. Recent work encompasses applications to climate adaptation, digital transformation, information regulation, rural development, regional integration, and policy evaluation.

COLLABORATION IDEAS

- Determinants of AI technology adoption in business, trade, and education in the context of choice uncertainty and imperfect information processing (e.g., limited attention);
- Impacts of AI-driven information systems on consumer choice behavior and strategic seller-buyer interactions (e.g., distinguishing intensive vs. extensive margin of information processing);
- Use of ML algorithms to solve complex grouping problems in economics (e.g., optimal composition of customs unions, trade blocks).



Thiago França NAVES

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■ BRAZIL

BIOGRAPHY

Dr. Thiago Naves holds a Ph.D. in Artificial Intelligence and serves as Head of R&D&I Research at the Federal University of Technology – Paraná (UTFPR), where he founded LAMIA – Laboratory of Machine Learning Applied to Industry and HUBX – Deep Tech Paraná, in partnership with the Government of the State of Paraná. He has secured more than BRL 80 million in funding from companies for the development of Artificial Intelligence products and coordinates an ecosystem of more than 300 scholarship-funded researchers. He serves as an EMBRAPPII Project Coordinator at CEIA – Center of Excellence in Artificial Intelligence. He is also a partner and advisor to startups and technology companies, specializing in public-private partnerships for AI-driven innovation. He has established academic recognition in the field of AI, with a strong track record in applied research, scientific production, researcher training, and technology transfer to the productive sector.

RESEARCH INTERESTS

His research focuses on Artificial Intelligence applied to high-impact technological solutions, with an emphasis on machine learning, computer vision, natural language processing, foundation models, and intelligent systems. He works on the research, development, and technology transfer of AI solutions for the industrial, healthcare, agribusiness, logistics, and services sectors, integrating applied research, innovation, and collaboration among academia, government, and industry.

COLLABORATION IDEAS

He is interested in establishing international collaborations in research and development focused on trustworthy and responsible Artificial Intelligence, particularly in the areas of synthetic data, foundation models, multimodal AI, and intelligent systems applied to industrial and societal challenges.



John Mulwa NDISYA

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■ GERMANY

BIOGRAPHY

John Mulwa Ndisya is an agricultural engineer and postdoctoral fellow at the Leibniz Institute for Agricultural Engineering and Bioeconomy in Potsdam, Germany. His research focuses on agricultural data infrastructure for AI adoption in agri-food systems, particularly where fragmented public and private data limit access, integration and reuse. Within the European Partnership Agriculture of Data, he coordinates a work package in which agri-food use cases involving different AI applications are prepared for sharing with relevant users across the European agricultural sector. He is also the lead researcher in the Renewable Energy for Food Security project, which develops modular AI-enabled food-processing units for decentralised operating environments.

RESEARCH INTERESTS

- 1. Shared and AI-ready agricultural data, federated data infrastructures and interoperability.
- 2. Applied AI for food and agricultural systems, with a focus on agri-food SMEs.
- 3. Embedded and edge AI systems integrating sensors, microcontrollers, actuators and lightweight machine-learning models for resource-constrained settings.

COLLABORATION IDEAS

- 1. Explore common standards, metadata and provenance requirements for sensor-generated agricultural data in Brazilian and German use cases.
- 2. Investigate federated AI approaches for joint model development and cross-regional validation without data centralisation
- 3. Discuss approaches for transferring and adapting data resources and AI technologies across comparable agri-food SMEs operating in different geographical contexts.



Rodrigo NEJM

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■ BRAZIL

BIOGRAPHY

PhD in Psychology and Digital Education Specialist at the Alana Institute.

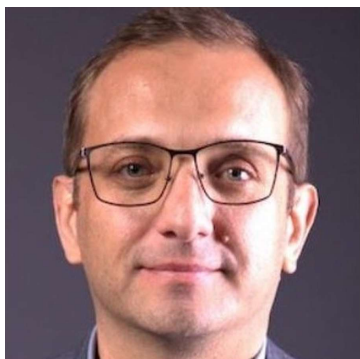
He completed a postdoctoral fellowship at the Graduate Program in Psychology at the Federal University of Bahia (UFBA). He holds a Master's degree in Social Management and Development from CIAGS/UFBA and a Bachelor's degree in Psychology from UNESP-Assis. With 20 years of experience as a social psychologist, his work focuses on child rights in the digital environment, digital well-being, digital citizenship education and child participation. From 2007 to 2023, he served as Education Director at SaferNet Brazil, where he coordinated national programs for the protection of children in digital environments. Member of the Tech Coalition Safe Online Research Fund Advisory Group (2020-2023). Since 2012, he has been a member of the Expert Group for the ICT Kids Online and ICT in Education surveys by Cetic.br, and a member of the Research Group on Interaction, Digital Technologies, and Society (GITS/UFBA).

RESEARCH INTERESTS

Children interpersonal relationships mediated by digital technologies, digital self and self-disclosure. Child rights by design approach, AI and child rights, AI and education.

COLLABORATION IDEAS

Rights by Design in AI applied to Education, digital education and child rights impact assessment



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■ BRAZIL

BIOGRAPHY

Professor of Technology and Innovation Management in the Department of Production Engineering at the São Carlos School of Engineering (EESC), University of São Paulo (USP). Director of the Center for Innovation and Sustainability Management (CEGIS) – www.cegis.com.br. Author of the first Brazilian book on roadmapping, *Roadmapping: A Strategic Approach to Innovation Management in Products, Services, and Technologies*. More information is available at www.roadmapping.com.br. He holds a PhD and a Master's degree in Production Engineering, as well as a Bachelor's degree in Mechanical Engineering, with an emphasis on product design, from EESC-USP. He completed a doctoral research period abroad and a postdoctoral fellowship at the Centre for Technology Management (CTM), Institute for Manufacturing (IfM), University of Cambridge, UK. He is also a specialist in Artificial Intelligence and Big Data from USP's Institute of Mathematics and Computer Science (ICMC).

RESEARCH INTERESTS

Strategic technology and innovation management, roadmapping, digitalization and digital technologies, and product and service development.

COLLABORATION IDEAS

Application of Digital Technologies in Technology and Innovation Management.



Juliani Chico Piai **PAIVA**

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■ BRAZIL

BIOGRAPHY

Juliani Chico Piai Paiva received the bachelor's and master's degrees in electrical engineering from the State University of Londrina in 2006 and 2009, respectively, and the Ph.D. degree in energy systems planning from the State University of Campinas in 2019. She is currently an Adjunct Professor and the Coordinator of the Master's Program in Electrical Engineering at the State University of Londrina (UEL). Her doctoral dissertation focused on energy poverty and affordability in Brazil. Her research, published in journals, such as Energy and Buildings and Utilities Policy, centers on Energy Poverty, Sustainable Development, Energy Efficiency, and Renewable Sources. She leads the Laboratory of Energy and Sustainable Development (LEDS) Research Group at UEL.

RESEARCH INTERESTS

Energy poverty; Energy planning; Sustainable development; Renewable energy; Energy efficiency.

COLLABORATION IDEAS

IA on energy planning and renewable.



Lara PICCOLO

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■ GERMANY

BIOGRAPHY

Lara is a researcher and educator in Human-Computer Interaction (HCI), with a focus on Inclusive and Socially Aware Design. As a Professor of Interaction Design at the CODE University of Applied Sciences in Berlin, Lara coordinates and teaches courses in Digital Design and Innovation, working at the intersection of design, computer science, and entrepreneurship. Lara serves as Chair of the International Federation for Information Processing (IFIP) Working Group on HCI Education, TC 13 - Human-Computer Interaction, leading an international collaboration to enhance HCI education globally. Lara was a research fellow at the Knowledge Media Institute of the Open University in the UK from 2014 to 2023. She received a PhD in Computer Science in 2015, with a focus on Human-Computer Interaction, from the Institute of Computing (IC) at UNICAMP in Brazil, where she was born and raised and where she began her research career.

RESEARCH INTERESTS

Lara's research is grounded in Socially-Aware Design, combining participatory techniques with methods to address technology's effects on society, enhancing inclusion, diversity, and accessibility. She is currently researching methodologies and tools to facilitate teamwork in inclusive workplaces for people with disabilities and those who are neurodiverse. Lara led and collaborated on international research projects spanning climate change mitigation, misinformation detection, gender-based violence prevention, and online abuse protection for children.

COLLABORATION IDEAS

Exploring the design of AI-based tools to support people with disabilities, neurodiverse individuals, and communities of minorities in Germany and Brazil. It investigates accessible, culturally grounded AI literacy frameworks that empower marginalised groups in education and work environments. Particularly interested in using AI to promote native language learning within Amazon communities in Brazil. The work bridges disability inclusion, neurodiversity, and linguistic heritage in a unified, participatory design approach.



Livy REAL

AI Scientist

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■ BRAZIL

BIOGRAPHY

Livy Real is a researcher in the fields of Natural Language Processing (NLP) and Artificial Intelligence, working at the intersection of industry and academia, as well as linguistics and computer science. She is currently an AI Scientist at the Kunumi Institute and the Institute of Computing at the Federal University of Amazonas (UFAM).

She holds a Ph.D. in Linguistics (UFPR/LaBRI). Her main research interests include the evaluation of large language models, natural language processing for low-resource languages, data sovereignty, AI for social good, and the development of responsible AI technologies.

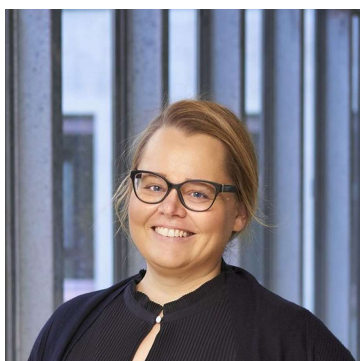
Since 2018, she has organized **OpenCor**, the Latin American Workshop on Open Data and Open Corpora. She is a member of the board of **Brasileiras em PLN** and, in 2026, serves on the organizing committee of **LAAI-Ethics**, the Latin American Conference on Ethics in Artificial Intelligence.

RESEARCH INTERESTS

Livy Real's research focuses on the evaluation of large language models, multilingual and low-resource NLP, responsible AI, data sovereignty, and culturally grounded AI safety. Her work bridges Natural Language Processing, Artificial Intelligence, and Linguistics, with an emphasis on developing trustworthy and socially relevant language technologies.

COLLABORATION IDEAS

Livy Real is interested in collaborations on multilingual and low-resource NLP, evaluation and alignment of large language models, AI safety, responsible AI, and culturally grounded approaches to AI governance. She is particularly interested in interdisciplinary collaborations that connect language technologies with ethics, public policy, and real-world applications, especially in Latin America and the Global South.



Evelyn REINMUTH

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■ GERMANY

BIOGRAPHY

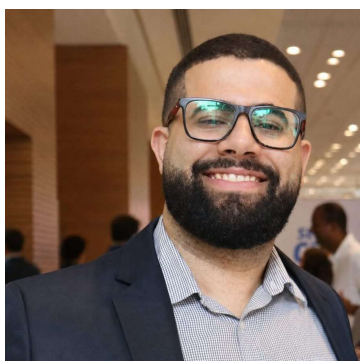
Dr Evelyn Reinmuth is the head of the Bioeconomy Office at the University of Hohenheim. She drives the development of a sustainable, circular bioeconomy through interdisciplinary collaboration and multi-stakeholder initiatives. She coordinates key activities of the European Bioeconomy University Alliance and conducts research in the Plant-Based Foods and Bio-based Resources in the Bioeconomy departments. Her research focuses on modelling integrated bioeconomic systems, with a particular focus on analysing the quality of bio-based feedstocks and intermediates, especially those derived from side streams, as well as strategies for the sustainable industrial use of biomass. She also identifies training needs and develops policy-relevant recommendations for action. She participates in the BMW's "Industrial Bioeconomy" dialogue platform and takes conceptual leadership and responsibility for knowledge transfer at the University of Hohenheim's "Agri-Food-Health Research Hub". She also coordinates the "S-TEC Center for Biointelligent Value Creation Hohenheim". She is also involved in setting up the C.A.T.A.L.I.S.T. platform for cellular agriculture. Embedded in Stuttgart's bioeconomy innovation ecosystem, this platform is breaking new ground in food production and technology transfer.

RESEARCH INTERESTS

My research focuses on modelling integrated bioeconomic systems, particularly assessing the quality of bio-based feedstocks from side streams and their sustainable industrial use. I am especially interested in interdisciplinary, cross-sectoral collaborations linking science, industry, and policy to scale sustainable bioeconomy solutions through standardization, machine learning, and the translation of biological knowledge into data-driven models.

COLLABORATION IDEAS

- Creating new datasets by systematically assessing data on bio-based resources.
- Advancing the standardization of biological resource use for industrial applications.



Anderson Rafael Correia Buarque da SILVA

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■ BRAZIL

BIOGRAPHY

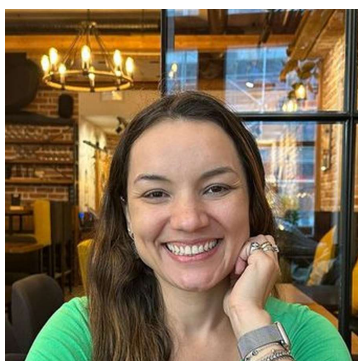
Ph.D. in Physics and Senior Researcher at SENAI CIMATEC, also serving as Deputy Coordinator of the EMBRAPPII Competence Center for Quantum Technologies. Research focuses on quantum algorithms for the simulation of quantum systems, with an emphasis on quantum state transport properties and quantum entanglement. Expertise includes quantum information, condensed matter physics, quantum algorithms, and quantum computing.

RESEARCH INTERESTS

Research interests include quantum machine learning, variational algorithms, applications of quantum computing, quantum cryptography, and immersive technologies for quantum education and the development of quantum technologies.

COLLABORATION IDEAS

Research interests also include artificial intelligence tools for immersive technologies, quantum machine learning, and other emerging applications at the intersection of artificial intelligence and quantum technologies.



Nádia Félix Felipe da SILVA

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■ BRAZIL

BIOGRAPHY

Dr. Nádia Félix Felipe da Silva is a Brazilian computer scientist and researcher specializing in Artificial Intelligence, with a strong focus on Natural Language Processing. She holds a Ph.D. and a postdoctoral degree in Computer Science and Computational Mathematics from the University of São Paulo (USP). Throughout her academic career, she has contributed to research in sentiment analysis, text mining, conversational AI, and language technologies for Portuguese. Dr. Silva has served as a professor at the Federal University of Goiás (UFG) and is currently a faculty member at the Federal University of Catalão (UFCAT). She has also collaborated with the Center of Excellence in Artificial Intelligence (CEIA) and as a visiting researcher at the University of Campinas (UNICAMP). Her work has been published in leading international journals and conferences, reflecting her commitment to advancing AI research and developing innovative language technologies for real-world applications.

RESEARCH INTERESTS

Natural Language Processing (NLP), Machine Learning, Large Language Models (LLMs), Sentiment Analysis, Named Entity Recognition (NER), Text Classification, Legal Natural Language Processing (Legal NLP), Conversational AI and Intelligent Agents, Portuguese Language Technologies, Representation Learning, Social Media Text Analysis, AI for Healthcare and Mental Health Text Analysis, Retrieval-Augmented Generation (RAG), AI Applications in Agriculture and Environmental Monitoring

COLLABORATION IDEAS

My research aligns with BRAGFOST 2026 through collaborations on Trustworthy AI, Natural Language Processing, Large Language Models (LLMs), and Generative AI. I am interested in responsible and multilingual AI, including Retrieval-Augmented Generation (RAG), explainable AI, synthetic data, and applications in healthcare, legal informatics, education, and sustainable agriculture. These collaborations aim to advance ethical AI and strengthen Brazil–Germany research partnerships.



Alyson Matheus de Carvalho **SOUZA**

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■ BRAZIL

BIOGRAPHY

Alyson Souza is an Associate Professor and researcher at the Digital Metropolis Institute (IMD/UFRN), where he specializes in the application of immersive technologies within the fields of arts and innovation. He serves as the coordinator of the Advanced Knowledge Center in Immersive Technologies (AKCIT) at IMD and leads the Adaptive Realities Lab at UFRN. Currently, he heads the project "Investigation of Immersive Technologies for the Development of New Experiences in the Arts," an initiative supported by AKCIT focused on pioneering new creative paradigms in Virtual and Mixed Reality. With a distinguished history of interventions at the intersection of aesthetics and technology, Souza has developed some of the most successful immersive exhibitions in the region, including MedusoZoa, Peixes do Semiárido, and Navegar Impreciso.

RESEARCH INTERESTS

Immersive Technologies, Virtual Reality, Arts, Musical Interfaces, Digitally Augmented Performances, Cultural Heritage

COLLABORATION IDEAS

Immersive technologies form a highly transversal field with a broad range of applications, which serve as the primary focus of my work. Having previously developed solutions for paleontology, nutrition, museums, education, professional training, serious games, and live performances, I am dedicated to exploring novel uses for these tools. My goal is to make immersive experiences more accessible to the general public, bridging the gap for a technology that remains relatively restrictive today.



Maximilian SPRANG

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■ GERMANY

BIOGRAPHY

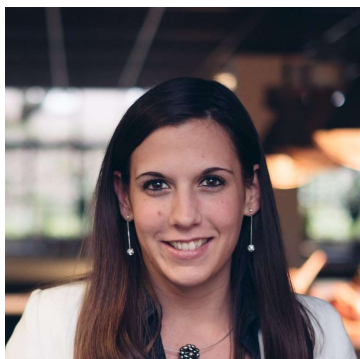
Dr. Maximilian Sprang is a junior group leader and lead bioinformatician at the University Medical Center of Johannes Gutenberg University Mainz, Germany. He studied Biology and Applied Bioinformatics in Mainz and completed his PhD (summa cum laude) in 2024 on machine-learning approaches to data quality in multi-omics, work recognised by the Boehringer Ingelheim Foundation Dissertation Award and the 2025 Einstein Early Career Researcher Award. His research develops generative and foundation models for biological sequences, together with methods to assess and safeguard against their dual-use and biosecurity risks. From 2027 he will become the first IQCB Scientist at the Institute for Quantitative and Computational Biology in Mainz. He previously held a role at the German Research Center for Artificial Intelligence (DFKI) and contributes to international open-science efforts including OpenBioML.

RESEARCH INTERESTS

I develop machine learning for functional genomics, focusing on generative and foundation models for biological sequence design (DNA, phage genomes, peptides). A second strand addresses the safety and security of these models: benchmarking dual-use and biosecurity risks and building safeguards for generative sequence design. I also work on data quality and reproducibility in multi-omics and on biologically informed ('visible') neural networks for multi-omics integration.

COLLABORATION IDEAS

I am looking for partners for safe-by-design genomic foundation models: co-developing risk benchmarks and safeguards for generative sequence design and shared evaluation frameworks. I am especially keen to work with Brazilian groups on pandemic surveillance, and on genomic foundation models for cancer-risk detection, biosecurity governance and AIxBio education, plus access to complementary datasets and compute.



Susan STEAD

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■ GERMANY

BIOGRAPHY

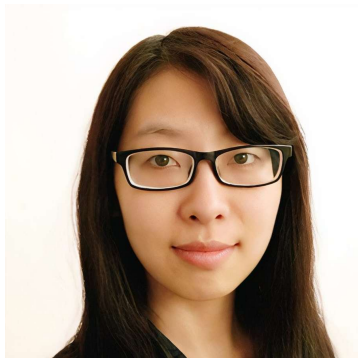
Dr. Susan Stead is an Assistant Professor at the Institute for Technology and Innovation Management (TIM) at RWTH Aachen University, where she leads the Healthcare Innovation Lab and the Digital Responsibility Lab. Her research focuses on technology and innovation management, digital transformation, and responsible innovation, with a particular emphasis on how organizations—and especially SMEs—can strategically adopt AI to create sustainable value. She investigates the managerial, organizational, and governance challenges of AI adoption, highlighting the importance of responsible innovation, trustworthy AI, and AI-supported decision-making. Through interdisciplinary research and collaboration with academia, industry, and public partners, her work promotes inclusive and AI-enabled business transformation, helping bridge the innovation gap for SMEs and strengthen long-term resilience.

RESEARCH INTERESTS

My research examines how AI-enabled innovation emerges through socio-technical interactions between technology, organizations, and diverse stakeholders. I am particularly interested in multi-actor collaboration, evolving professional and organizational role identities, and the governance of responsible AI adoption. By studying how organizations navigate digital transformation within innovation ecosystems, I explore pathways toward inclusive, trustworthy, and sustainable AI-enabled innovation.

COLLABORATION IDEAS

I seek collaborations on AI-enabled innovation, digital transformation, and responsible technology governance, particularly for SMEs. I am interested in developing joint research projects, comparative international studies, and co-authored publications that connect technology and innovation management with engineering, social sciences, and policy. Together, we can advance evidence-based approaches for trustworthy AI and resilient innovation ecosystems. So many options and I'm really excited to exchange.



Yao SUN

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■ GERMANY

BIOGRAPHY

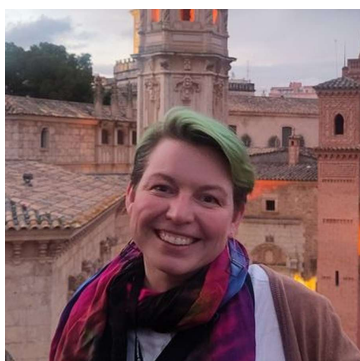
Dr.-Ing. Yao Sun is a research associate at the German Aerospace Center (DLR). Her research focuses on geospatial artificial intelligence, remote sensing, geoinformatics, and multimodal data integration for disaster-risk and urban-resilience applications. She works with satellite imagery, street-level imagery, geotext, large language models, and vision-language models to extract building-related information, assess exposure and vulnerability, and support decision-making in high-stakes disaster-management contexts. Her current work explores vision-language models and addresses trustworthy and explainable AI for building vulnerability assessment, as well as synthetic data generation and benchmark development for responsible geospatial AI.

RESEARCH INTERESTS

Geospatial AI; remote sensing data analysis; street-level imagery; vision-language models; multimodal urban data fusion; building exposure and vulnerability; disaster risk reduction; explainable AI for high-stakes geospatial decision support.

COLLABORATION IDEAS

I am interested in Brazil-Germany collaborations that use geospatial data and AI to address urban and environmental challenges. Potential topics include combining satellite Earth observation, street-level imagery, maps, text and local knowledge for studying urban growth, informal settlements such as Brazilian favelas, environmental exposure, disaster risk and public-sector decision support. I am particularly interested in identifying shared research questions where remote sensing and multimodal data can complement local expertise and field-based knowledge.



Sabine WEBER

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■ GERMANY

BIOGRAPHY

Dr. Sabine Weber is a postdoctoral researcher at the Bamberg NLP Group, specializing in subjective NLP tasks and social meaning. They earned their Ph.D. in Computer Science from the University of Edinburgh, focusing on multilingual entailment detection, language adaptation and knowledge graphs. Before that they received a B.Sc. and a M.Sc. in Computer Engineering from the TU Berlin, and a B.A. in Slavic Languages from the Humboldt University Berlin.

In addition to building models that are more robust and flexible, they are interested in making NLP models more explainable and examining the societal impact of NLP. They are also a core organizer for the volunteer-run organization Queer in AI.

RESEARCH INTERESTS

Language is inherently social, which is why I am interested in incorporating social dimensions into NLP, for example through metadata. My research focuses on subjective tasks such as emotion classification, offensiveness detection, and humor recognition, which naturally intersect with questions of societal bias in NLP models. In particular, I am interested in how these models process and represent in-group and out-group language.

COLLABORATION IDEAS

I would love to work on the creation and analysis of text corpora in subjective NLP tasks that span different languages and cultures, for example to create better culturally aware models of emotion in text and improve the robustness and fairness of NLP systems. More broadly, I am interested in understanding how social and cultural context shapes language use and how these factors can be incorporated into models to better reflect the diversity of human communication.



Ingrid WINKLER

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Advanced Knowledge Center for Immersive Technologies (AKCIT)

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■ BRAZIL

BIOGRAPHY

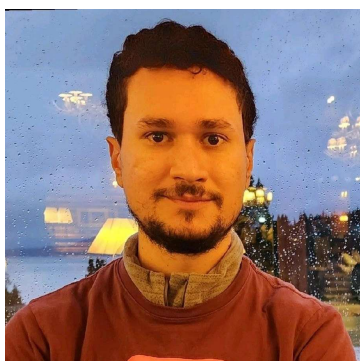
Dr. Ingrid Winkler's research explores the potential of immersive technologies to drive a more sustainable and human-centric Industry 5.0. She is the Vice-Chair of the Special Committee on VR at the Brazilian Computing Society (SBC) and holds a DT-1D Research Fellowship awarded by Brazil's National Council for Scientific and Technological Development (CNPq). At SENAI CIMATEC University, Dr. Winkler leads the Institute for Science, Innovation, and Technology in Industry 4.0 (INCITE INDUSTRIA 4.0). She has held international research positions as a Visiting Researcher at Nanyang Technological University (NTU, Singapore) and is currently affiliated with Lusíada University (Portugal) and the EMBRAPA Advanced Knowledge Center for Immersive Technologies (AKCIT/ UFG). Dr. Winkler has coordinated more than 50 R&D&I projects in collaboration with industry partners like EMBRAER, FORD, Great Wall Motors, NVIDIA, Petrobras, SHELL, HP, Fiocruz, and Continental Mabor Portugal.

RESEARCH INTERESTS

immersive technologies; virtual reality; extended reality; Industry 5.0; sustainable manufacturing; human-centric production; agentic AI

COLLABORATION IDEAS

Collaborating on research at the intersection of immersive technologies and agentic AI for a sustainable and human-centric Industry 5.0, including adaptive immersive environments for industrial training, worker safety, ergonomic assessment, decision support, and immersive digital twins. The collaborations could include to develop and evaluate solutions that improve operational efficiency while prioritizing worker well-being, inclusion and environmental sustainability



Otávio Calaça XAVIER

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■ BRAZIL

BIOGRAPHY

Otávio Calaça Xavier holds a Ph.D. and a Master's degree in Computer Science from the Federal University of Goiás (UFG). He teaches undergraduate and graduate courses in Artificial Intelligence, Software Engineering, and Information Systems at UFG and the Federal Institute of Goiás (IFG). He has been a researcher at CEIA (Center of Excellence in Artificial Intelligence) since its founding and is a co-founder of Deep Learning Brasil. His research focuses on graph neural networks, knowledge graphs, information retrieval, retrieval-augmented generation (RAG), and neuro-symbolic AI. He has more than 22 years of experience in software and data engineering, including 16 years applying machine learning to production systems, with experience at both Brazilian and international companies. Since 2007, he has delivered more than 100 talks at technology events across Brazil and contributes to open-source projects, including PyTorch Geometric.

RESEARCH INTERESTS

Neuro-Symbolic AI; Retrieval-Augmented Generation (RAG); Knowledge Graphs; Graph Neural Networks (GNNs); Information Retrieval and Hybrid Search; Multi-hop Reasoning; Multi-Agent Systems with LLMs; Temporal and Dynamic Graphs; Evaluation and Observability of Generative Systems; Interpretability and Responsible AI.

COLLABORATION IDEAS

I am interested in Brazil-Germany collaborations in trustworthy and responsible Generative AI. At CEIA/UFG, IFG, and in industry, I work with RAG, knowledge graphs, and neuro-symbolic approaches to reduce hallucinations and enhance the explainability of LLMs. I see opportunities for multilingual benchmarks and applications in education, research, and innovation for SMEs. As potential outcomes, I propose joint research projects and publications, researcher exchanges, and academic visits with scientific and socioeconomic impact..



Zelun Tony ZHANG

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■ GERMANY

BIOGRAPHY

Tony Zhang is a postdoctoral researcher at the Chair of Human-Centered Technologies for Learning at the Technical University of Munich. He holds a Dr. rer. nat. in Media Informatics from LMU Munich, supervised by Prof. Dr. Andreas Butz and Prof. Dr. Heinrich Hußmann. He completed his doctoral degree while working as researcher in the Human-Centered Engineering team at fortiss GmbH. His dissertation was at the intersection of HCI and AI, investigating how AI—despite its imperfection—can be used to support human decision-making in high-stakes domains like aviation. He holds both a Master's and Bachelor's degree in Electrical Engineering from RWTH Aachen University. In between, he studied for one year as an exchange student at the Department of Computer Science and Technology at Tsinghua University in Beijing, China.

RESEARCH INTERESTS

My research interests revolve broadly around how AI can be used to augment rather than undermine human cognition, using methods from both HCI and machine learning.

Current research:

- Formative, dialogic, and synchronous AI writing feedback for elementary school students.
- AI tool that helps researchers read papers more efficiently, while maintaining cognitive engagement with the paper.

Planned research:

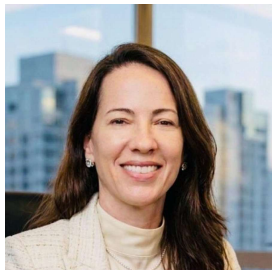
- How to evaluate reliance on GenAI?
- How to design process-oriented AI support?

COLLABORATION IDEAS

- Grounding of human-AI interaction design or evaluation in learning science or cognitive psychology.
- Process-oriented AI support in various real-world applications.
- LLM-finetuning for some cognitive augmentation objective.

Guests & staff

Representatives of CAPES, the Alexander von Humboldt Foundation and CEIA-UFG.



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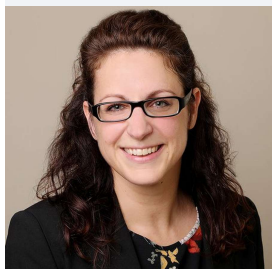
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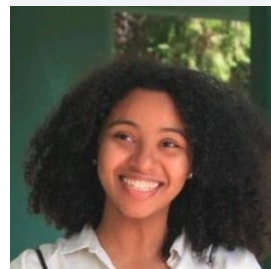
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Inka LOECK

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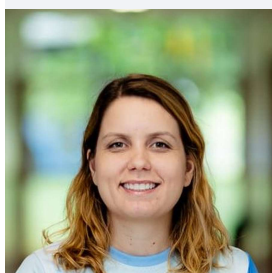
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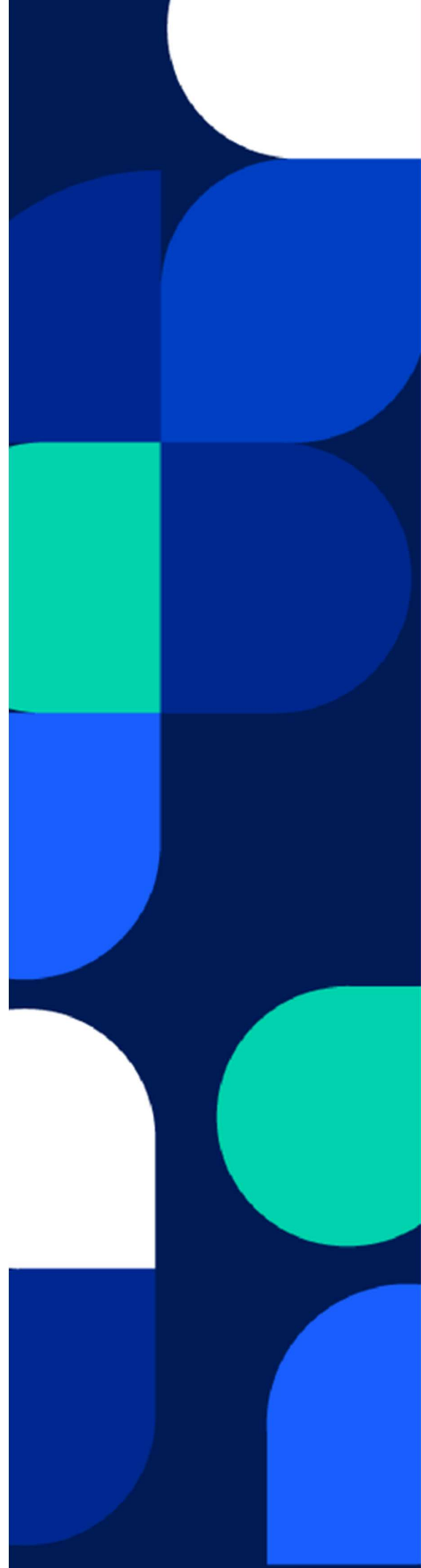
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PARTNER

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**CAPEX – Federal
Agency for Support
and Evaluation of
Graduate Education**

CAPES is a public foundation of the Brazilian Ministry of Education and one of the two organizations behind the BRAGFOST series.

BRAZILIAN FEDERAL AGENCY FOR SUPPORT AND EVALUATION OF GRADUATE EDUCATION (CAPES)

A Public Foundation of the Ministry of Education (MEC), Capes plays a fundamental role in the expansion and consolidation of the *stricto sensu* (master and doctorate levels) graduate courses in Brazil. Furthermore, it sponsors the training of basic education teachers and of highly qualified personnel abroad, thus contributing to the internationalization of Brazilian higher education.

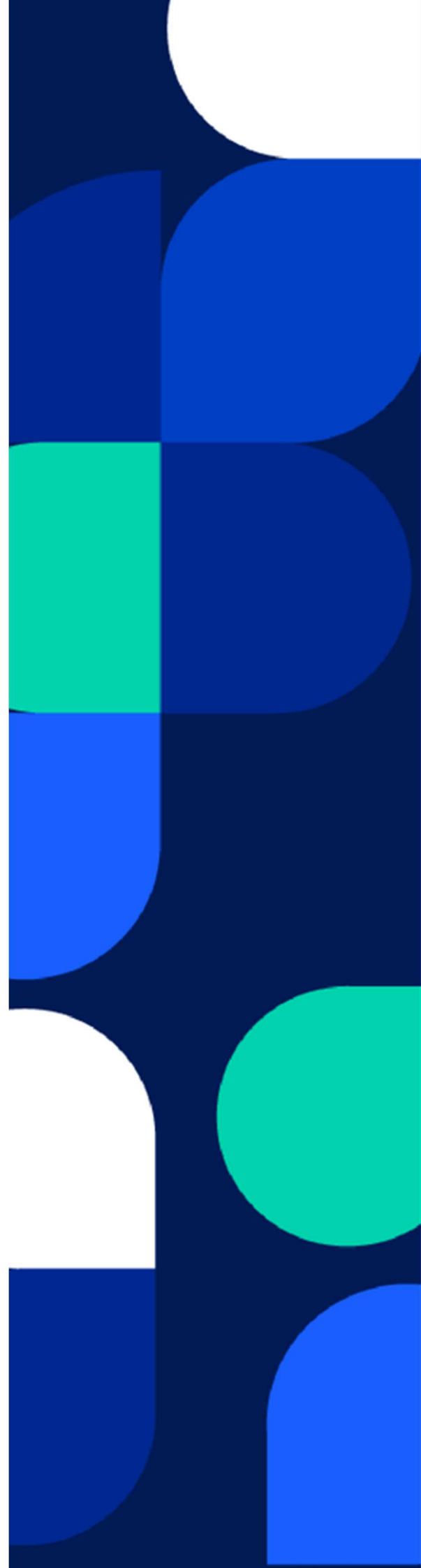
CAPES activities can be grouped in the following lines of action, each developed by a structured set of programs:

I	Postgraduate Evaluation The National Graduate System evaluation is conducted by Capes every 4 years with the participation of the academic-scientific community. Academic and professional postgraduate courses are graded with a 3 to 7 scale, being 3 the minimum required for a course to be considered recognized by Capes, therefore authorized to operate at a national level.
II	Scholarship Opportunities The domestic <i>stricto sensu</i> master and doctoral postgraduate scholarships from Capes are institutional, that is, they are managed by the Higher Education Institutions (HEI) and postgraduate courses, which are responsible for the selection and monitoring of the scholarship holders. The fellowships abroad are primarily directed towards doctorate level and postgraduate research. All grants are awarded through competitive selection processes and public calls, which are widely disclosed to the Academic Community.
III	Financial Grants Capes provides resources to graduate courses in order to foster projects regarded as strategic for the development of knowledge as Engineering, National Defense, Nanobiotechnology, Oceanography and Marine Sciences, Medical Research, Morphology, Gender Studies (with emphasis in studies on violence against women), Educational themes (including Special Needs issues), Bio-energy and Bio-fuels, among others.
IV	International Cooperation It is a strategic component of Capes' mission. Thus, it is directed to the scientific-technological capacity building towards the Excellency of Brazilian Higher Education. As a valuable component of modern higher education, internationalization is encouraged in a comprehensive way, not only through the mobility of students and faculty, but also the exchange of ideas, the integrating of an international dimension into the teaching, research and service functions of higher education institutions.
V	Broad Access to Scientific Content (Periodical Portal) It is a virtual library that provides access to international scientific production for education and research institutions in Brazil. It has a collection of thousands of titles with full text, reference bases, bases dedicated exclusively to patents, as well as books, encyclopedias and reference works, technical standards, statistics and audiovisual content. The Portal was created to overcome the lack of access of Brazilian libraries to international scientific

	information. It supports the growth of national scientific production as well as the Brazilian academic recognition abroad. Therefore, it is a fundamental tool within Capes mission, in the support and development of postgraduate education in Brazil.
VI	Teacher Training for Basic Education Training efforts are directed at supporting studies, research and innovation projects for strengthening the teaching profession. This action also counts on the Open University of Brazil (UAB), a system composed of public universities that offers higher education courses for the population that have difficulty accessing university education, with distance learning.

More information

Find more information at <http://www.capes.gov.br>



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**Alexander von
Humboldt
Foundation**

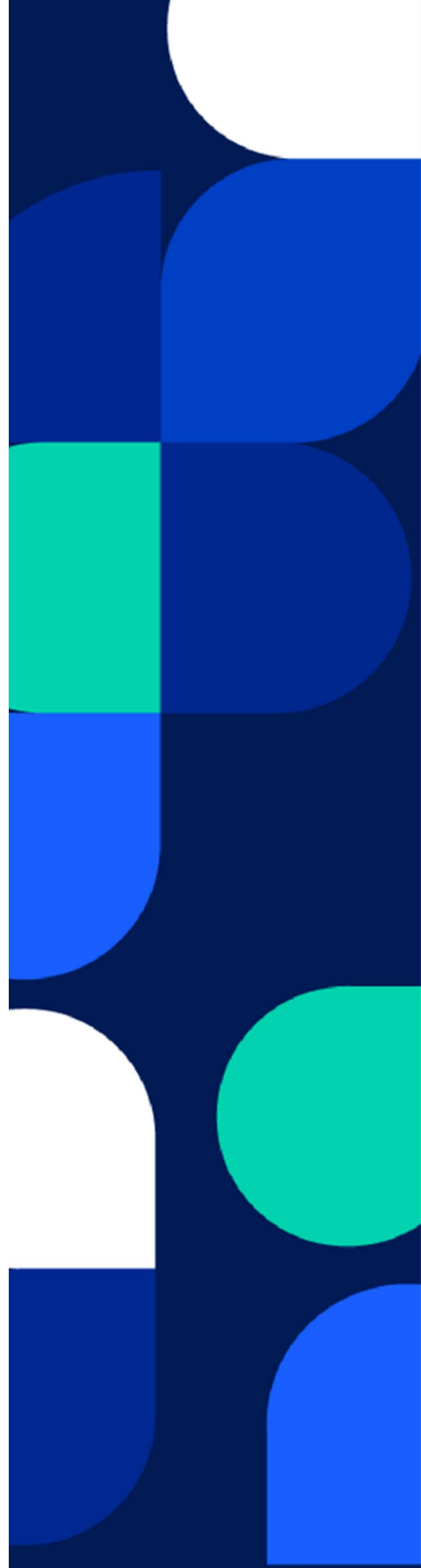
The Alexander von Humboldt (AvH) Foundation promotes academic cooperation between excellent scientists and scholars from abroad and from Germany. To this end it grants more than 700 research fellowships and research awards annually. These allow foreign scientists and scholars to conduct a research stay in Germany and to collaborate closely with their hosts and partners. Scientists or scholars from Germany can carry out a research project abroad as a guest of one of well over 30.000 Humboldt Foundation alumni around the globe – the Humboldtians. Nowadays the foundation's network embraces scientists and scholars from all disciplines in more than 140 countries - including 63 Nobel Laureates.

Each year, the Alexander von Humboldt Foundation grants:

- *Humboldt Research Fellowships* to highly qualified foreign scholars and scientists who have either completed their doctorates in the last four years (postdoctoral researchers) or less than twelve years ago (experienced researchers), enabling them to undertake long-term periods of research in Germany;
- *Feodor Lynen Research Fellowships* to highly qualified German scholars and scientists who have either completed their doctorates in the last four years (postdoctoral researchers) or less than twelve years ago (experienced researchers), enabling them to spend periods of research abroad;
- *International Climate Protection Fellowships* to prospective leaders and postdoctoral researchers with expertise in climate protection from non-European developing and transition countries, giving them the opportunity to carry out a 12–24-month research project in Germany;
- *The Henriette Herz Scouting Programme* grants for outstanding (junior) professors and (junior) research group leaders in Germany, allowing them to recommend up to three junior researchers from abroad for a Humboldt Research Fellowship that will be directly granted after formal approval;
- *Humboldt Research Awards* to internationally recognized foreign scholars and scientists (nominations by German scholars/scientists);
- *Friedrich Wilhelm Bessel Research Awards* to outstanding scholars and scientists from abroad who have completed their doctorates less than eighteen years ago;
- *Alexander von Humboldt Professorships* to top-rank researchers of all disciplines working abroad who are eligible to be appointed to full professorships in Germany to carry out ground-breaking and long-term research (nominations by German universities)
- *Philipp Schwartz Initiative* fellowships to researchers who face significant and ongoing personal threats in their countries of origin (outside the EU), enabling them to continue their work at German universities and research institutions for 24 months, with the possibility of a 12-month extension.

More information

For more information please refer to our website <http://www.humboldt-foundation.de> or contact our main office at <info@avh.de>.



WHAT COMES NEXT

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Follow-up program CONNECT

CONNECT Follow-up Program

The Frontiers of Research Symposia enable researchers to establish long-term bi-national cooperations that strengthen the scientific relationship between Germany and the respective partner country. Every Frontiers participant can apply to the dedicated follow-up program CONNECT to conduct working visits in the partner country for up to 30 days per conference to prepare joint research projects or articles.

Application requirements

The applicant has participated in a Brazilian-German Frontiers of Science and Technology (BRAGFOST) Symposium and seeks to further deepen cooperation with at least one participant of the partner country. Cooperating partners must have attended the same conference. Either the applicant or the hosting partner must be based in Germany. The applicant shall submit a short report (max. 3 pages) to the Alexander von Humboldt Foundation within 2 months after termination of the working visit.

Allowance rates

There are three different grades according to the position held by the applicant.

For working visits lasting **up to 22 days** the allowance is calculated on the basis of daily rates valid at the time of the application.

GRADE	ALLOWANCE	POSITION
Grade A	EUR 135 per day	Assistant Professor, Lecturer, Researcher, Junior Professor, Post-Doc
Grade B	EUR 162 per day	Associate Professor, Senior Lecturer, Reader
Grade C	EUR 189 per day	Full Professor, Head of Research Institute

For working visits lasting **longer than 22 days** a monthly lump sum is granted.

GRADE	ALLOWANCE	POSITION
Grade A	EUR 3.000 per month	Assistant Professor, Lecturer, Researcher, Junior Professor, Post-Doc
Grade B	EUR 3.600 per month	Associate Professor, Senior Lecturer, Reader
Grade C	EUR 4.200 per month	Full Professor, Head of Research Institute

The Alexander von Humboldt Foundation does not cover any additional costs (e.g. travel costs). Grants for applicants from Germany are transferred to their bank accounts in Germany or the SEPA zone whereas grants for applicants from a partner country are transferred either to a private bank account in the SEPA zone or to the host university's or institute's bank account in Germany, from where the allowance is paid to the applicant.

Online application

Further information on the online application is available at www.humboldt-foundation.de/connect

Please note the following points:

- Submit your online application well in advance of your planned working visit.
- Discuss all details (including the dates) of your planned working visit with your academic host before applying.

Complete applications include the following documents in English:

- application form
- curriculum vitae of the applicant
- description (3 to max. 5 pages) of the joint collaboration project
- invitation letter of the academic host
- list of key publications (max. one page)

Upon receipt of your application, the processing takes up to four weeks. Retrospective approval is not possible. The Alexander von Humboldt Foundation requires applicants to follow the Rules of Good Scientific Practice and the legally binding principles of scientific ethics.

Contact person for questions on the application process:

Stephanie Dill

Phone: + 49 (0)30 2045 45 26

Email: <stephanie.dill@avh.de>

CAPES · Alexander von Humboldt Foundation · CEIA-UFG
Goiânia, 24 – 27 September 2026

